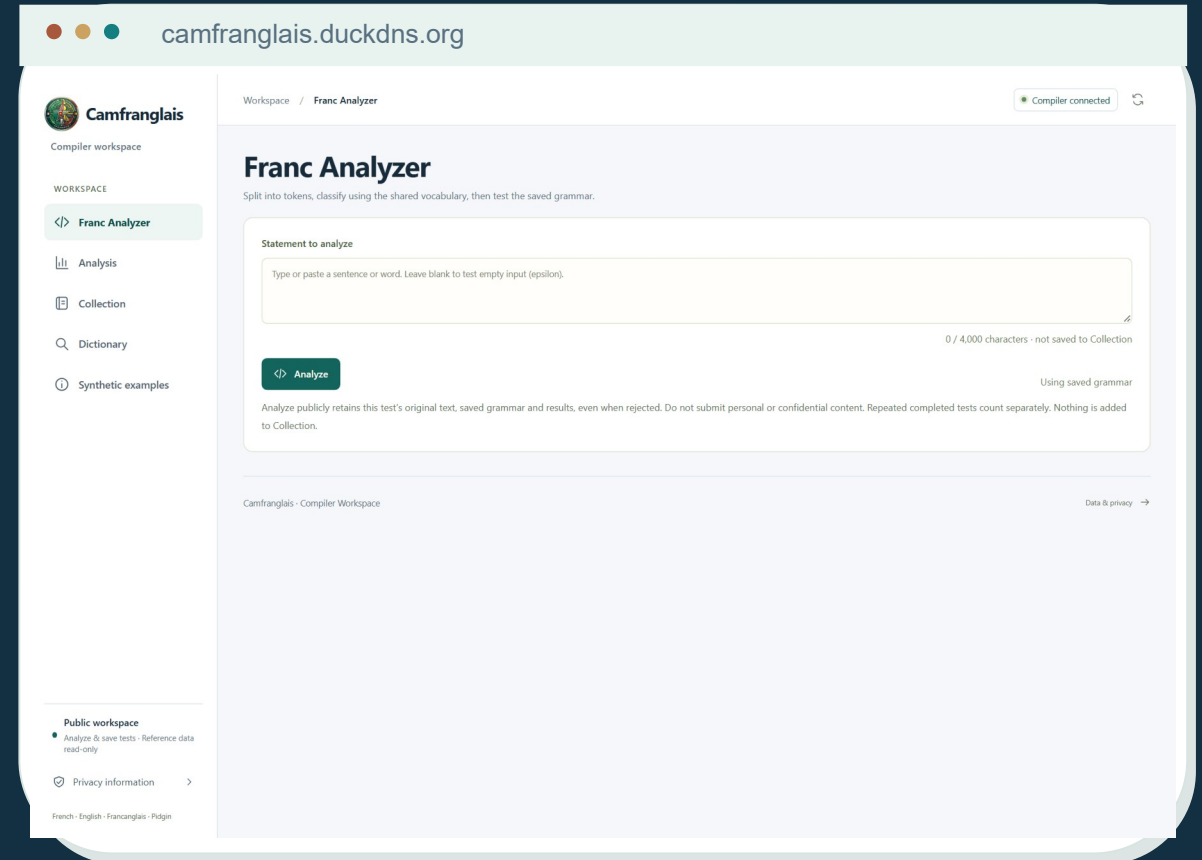


Camfranglais

From everyday language to
transparent compiler
decisions.

THE APP. THE RULES. THE FILES.

100-slide technical presentation
CS4110 / Summer 2026 / SET A



The group behind the project

Three members. One shared compiler workspace.

01

**Kwete Ngouba
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02

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03

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Three questions this deck answers

Start with the experience. Follow the data. Find the implementation.

01

USE

What can a visitor do on each of the five pages?

02

TRACE

How does a statement become tokens, verdicts and a saved result?

03

LOCATE

Which real source file owns each part of that behavior?

Mixed language. Traceable rules.

A teaching tool for inspecting informal urban communication.

Multiple influences

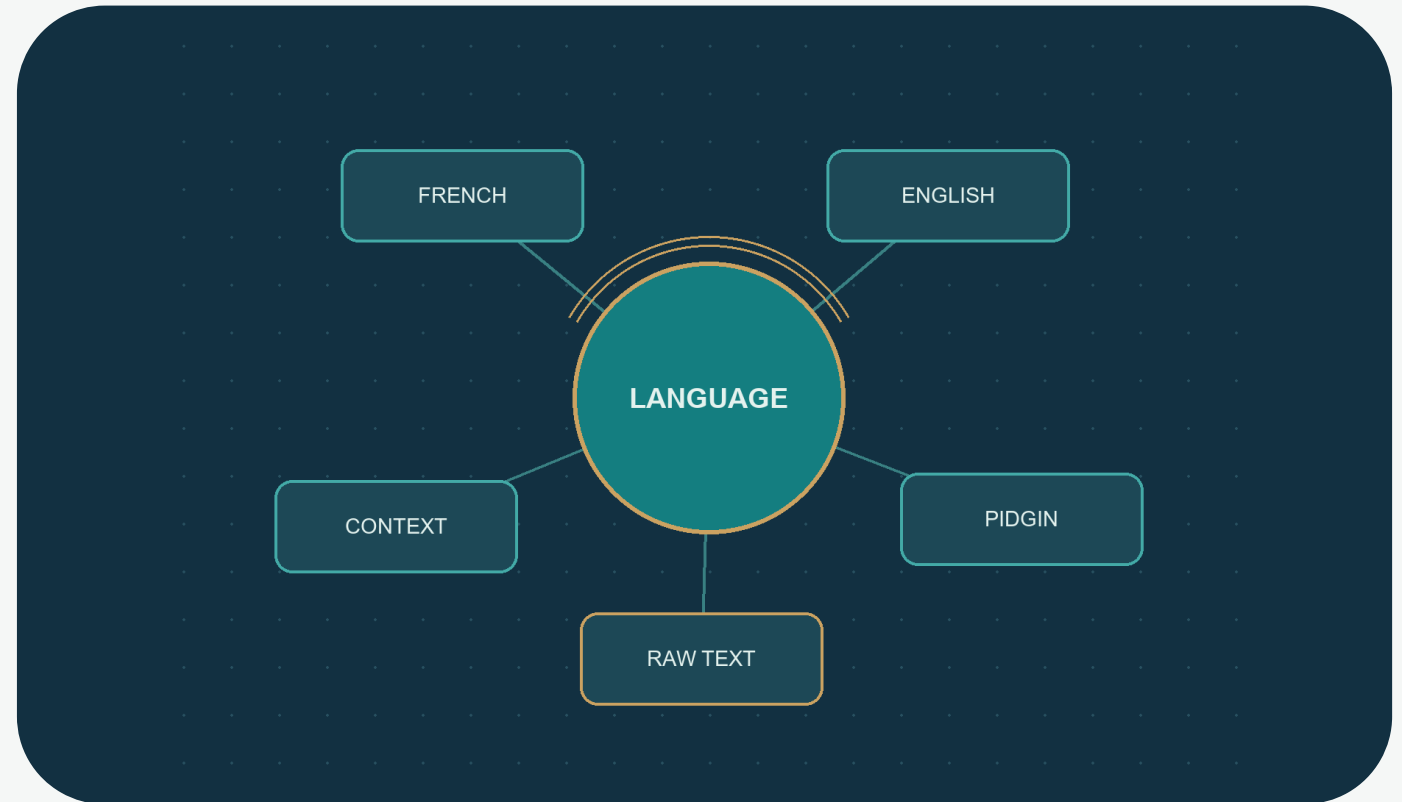
French, English, Camfranglais and Pidgin evidence can coexist.

Original wording

Keep spelling, apostrophes and punctuation visible.

Inspectable decisions

Explain categories and grammar, not a translation.



IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py

compiler/lexer/languages.py

Five pages, one connected workflow

Click a page card in the slideshow to jump to its walkthrough.

01

Franc Analyzer

Enter text and record a test.

02

Analysis

Inspect saved results and statistics.

03

Collection

Browse supplied records, context and audio.

04

Dictionary

Search source-labelled vocabulary.

05

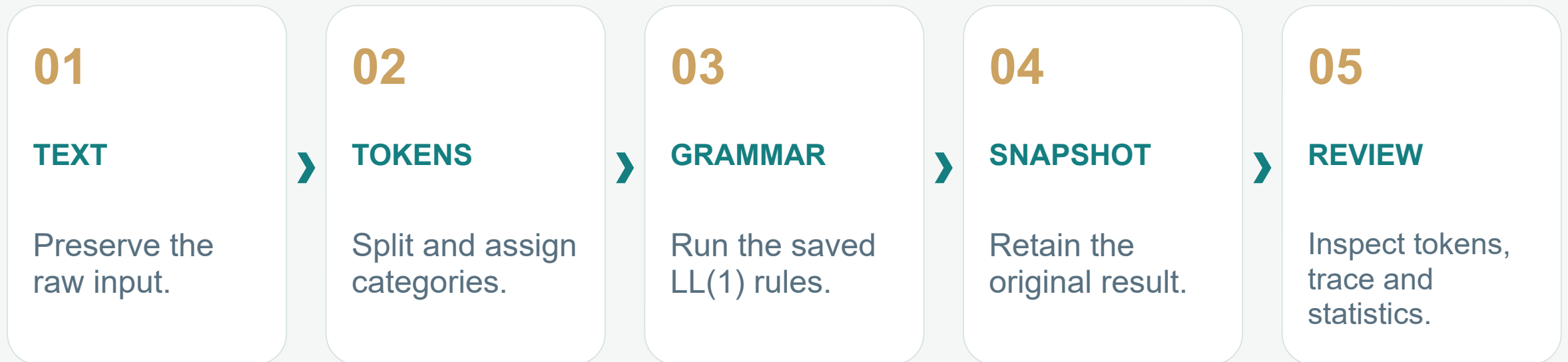
Synthetic examples

Explore clearly constructed practice material.

**ONE SHARED
PUBLIC WORKSPACE**

From statement to saved evidence

The main journey has five visible outcomes.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx
backend/analyzer_history.py
CS4110 / SETA / 27 SEP 2026

backend/analyzer.py

Two verdicts. Different questions.

A familiar word sequence is not automatically a grammatical match.

01

VOCABULARY

Are any tokens UNKNOWN?
No unknowns means vocabulary
approval.

02

GRAMMAR

Does the category sequence fit the
saved CFG?
Known words can still reject.

IMPLEMENTATION / EVIDENCE

frontend/src/VocabularyVerdict.tsx
compiler/parser/predictive.py
CS4110 / SETA / 27 SEP 2026

backend/analyzer_models.py

Public does not mean editable

The permission boundary is deliberate.

Action	Public behavior
Analyze and save tests	Allowed; results are retained publicly
Read saved tests	Allowed; original snapshots
Browse Collection and grammar	Read-only
Play existing recordings	Allowed when a matching file exists
Edit reference data or recordings	Denied
Use accounts or private maintenance APIs	Not exposed publicly

IMPLEMENTATION / EVIDENCE

backend/public_access.py

frontend/src/PublicSession.tsx

Every feature has a source

Read the source rail as a map, not just a list of filenames.

frontend/src

App.tsx

Chooses the page and connects navigation.

backend

analyzer_api.py

Receives analyzer HTTP requests.

compiler/lexer

tokenizer.py

Splits and classifies the input.

compiler/parser

predictive.py

Executes the bounded parsing stack.

IMPLEMENTATION / EVIDENCE

frontend/src/App.tsx
compiler/lexer/tokenizer.py
CS4110 / SETA / 27 SEP 2026

backend/analyzer_api.py
compiler/parser/predictive.py

The project at a glance

Different numbers describe different parts of the system.

5

Public pages

12

Statements in the controlled study

938

Reference dictionary entries

26

Constructed practice examples

IMPLEMENTATION / EVIDENCE

frontend/src/App.tsx
docs/README.md
CS4110 / SET A / 27 SEP 2026

docs/evidence/final-analysis-20260927.json

Start in Franc Analyzer

No account gate stands between the visitor and the input.

One main input

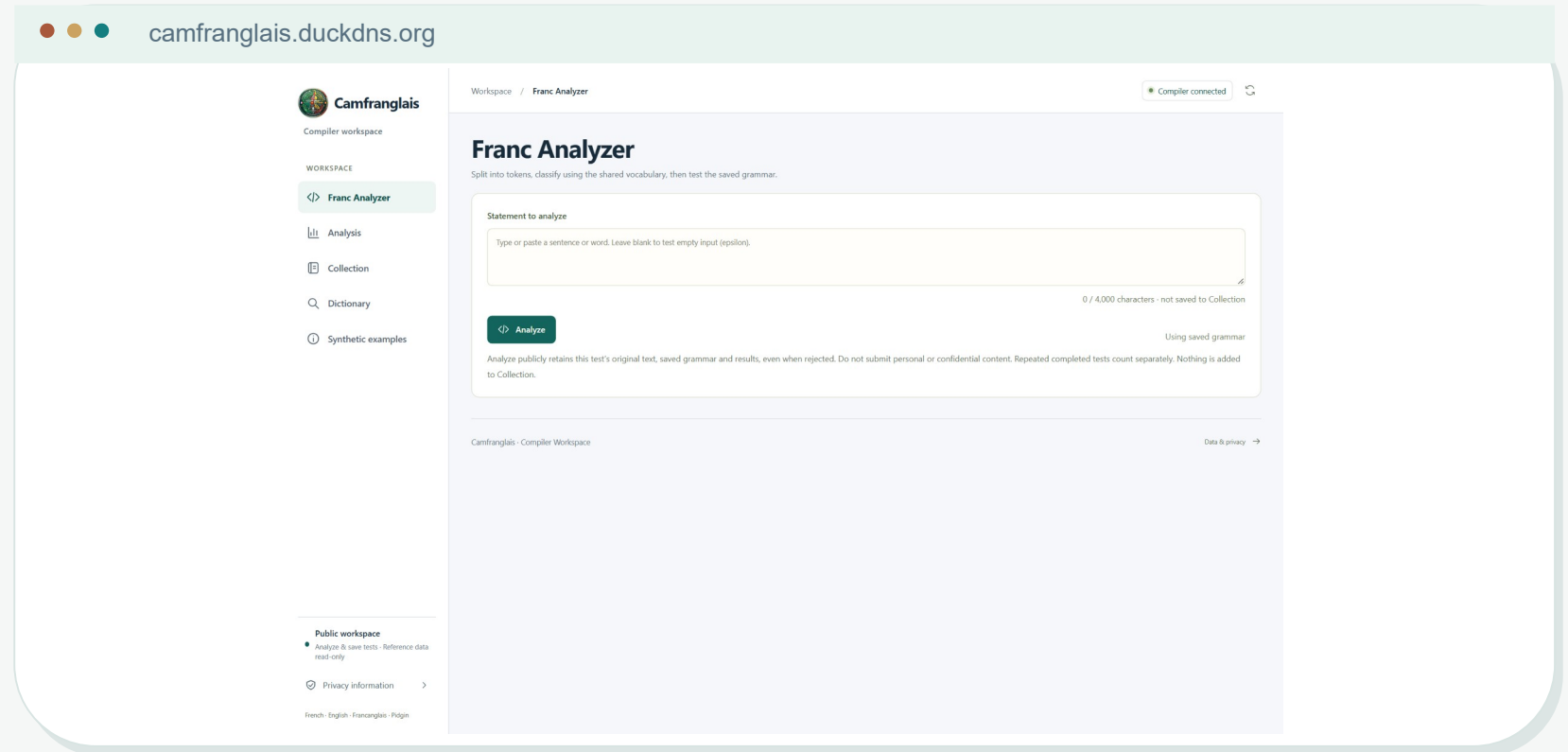
Type or paste a sentence or word.

Saved grammar

The public interface uses the operator-saved grammar.

Visible retention notice

The interface explains that tests become public.



REAL DEPLOYED APP / 27 SEP 2026 / crop only where noted

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx

frontend/src/PublicSession.tsx

Keep the speaker's original text

The input is evidence, not text to silently repair.

4,000 characters

The UI bounds one analyzer request.

No hidden correction

Keep case, spelling and punctuation.

Separate destination

A test is not added to Collection.

The screenshot shows a web browser window with the address bar displaying 'camfranglais.duckdns.org'. The page title is 'Franc Analyzer'. Below the title, a subtitle reads 'Split into tokens, classify using the shared vocabulary, then test the saved grammar.' The main content area features a text input field labeled 'Statement to analyze' with a placeholder text 'Type or paste a sentence or word. Leave blank to test empty input (epsilon)'. To the right of the input field, it shows '0 / 4,000 characters · not saved to Collection'. Below the input field is a green button with a code icon and the text 'Analyze'. At the bottom of the main content area, a disclaimer states: 'Analyze publicly retains this test's original text, saved grammar and results, even when rejected. Do not submit personal or confidential content. Repeated completed tests count separately. Nothing is added to Collection.' The footer of the page shows 'Camfranglais · Compiler Workspace' on the left and 'Data & privacy →' on the right.

REAL DEPLOYED APP / 27 SEP 2026 / crop only where noted

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx
backend/analyzer_models.py
CS4110 / SET A / 27 SEP 2026

frontend/src/types.ts

What happens after Analyze?

A click starts a reproducible, server-side workflow.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx
backend/analyzer_history.py
CS4110 / SETA / 27 SEP 2026

backend/analyzer_api.py
backend/shared_workspace.py

The client sends a reproducible request

Three fields connect the UI to a specific computation.

SOURCE WALKTHROUGH / ABBREVIATED WHERE MARKED

```
POST /api/analyzer/tests
```

```
{  
  "request_id": "<new or reused UUID>",  
  "text": "je suis kass je go nang",  
  "grammar": "<exact saved grammar>"  
}
```

Text

The original string, not a translation.

Grammar

The saved rule text used for this test.

Request ID

Identical retry can find the same result.

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx
backend/analyzer_history.py
CS4110 / SETA / 27 SEP 2026

backend/analyzer_models.py

Waiting, stopping and retrying

Browser state and server completion are not the same thing.

01

WAIT

The analyzer allows up to 60 seconds for its request.

02

STOP WAITING

Abort the local wait, not a guaranteed server rollback.

03

RETRY IDENTICALLY

Keep the request ID to recover the already-saved result.

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx

frontend/src/api.ts

CS4110 / SETA / 27 SEP 2026

frontend/src/useRequest.ts

A test is an immutable snapshot

Opening an old result does not run today's rules again.

SOURCE

The exact original text.

RULES

The grammar used at that time.

RESULT

Tokens, approval and parser trace.

IDENTITY

A stable test ID and recorded time.

IMPLEMENTATION / EVIDENCE

backend/analyzer_models.py
backend/shared_workspace.py
CS4110 / SETA / 27 SEP 2026

backend/analyzer_history.py

An accepted example, token by token

S02: je suis kass je go nang

je suis kass je go nang

je

FF

suis

VERB

kass

ADJECTIVE

je

FF

go

VERB

nang

VERB

6 tokens

Every raw word is retained.

No UNKNOWN

Vocabulary approval passes.

Accepted CFG

All six categories are consumed.

IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py
docs/evidence/final-analysis-20260927.json
CS4110 / SETA / 27 SEP 2026

compiler/parser/predictive.py

Rejection keeps the evidence

The two unresolved corpus forms are not erased or repaired.

01

S09 / n'ais

The sixth token is UNKNOWN.
Five tokens are consumed before
rejection.

02

S10 / alli

The sixth token is UNKNOWN.
Later tokens are not silently skipped.

IMPLEMENTATION / EVIDENCE

compiler/tests/yaounde_cases.py
docs/evidence/final-analysis-20260927.json
CS4110 / SET A / 27 SEP 2026

compiler/parser/predictive.py

Reference, then draft, then explicit Analyze

Opening a library item does not secretly create a test.

01

CHOOSE

Collection, Dictionary
or an example



02

HAND OFF

Copy its text to Franc
Analyzer



03

REVIEW

Inspect the draft and
retention notice



04

ANALYZE

Explicitly run and
save the test

FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

frontend/src/App.tsx
frontend/src/Dictionary.tsx
CS4110 / SETA / 27 SEP 2026

frontend/src/Collection.tsx
frontend/src/Examples.tsx

Analyzer: the files to open

Use this map when explaining or debugging the main workflow.

frontend/src

FrancAnalyzer.tsx

Input, saved grammar, submit, retry and selected test.

frontend/src

api.ts

JSON requests, CSRF headers and error messages.

backend

analyzer_api.py

Public analyzer endpoints.

backend

analyzer_history.py

Record, retrieve and report saved tests.

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx

backend/analyzer_api.py

CS4110 / SETA / 27 SEP 2026

frontend/src/api.ts

backend/analyzer_history.py

Analysis is the evidence desk

Statistics and original results from publicly saved tests.

Dashboard

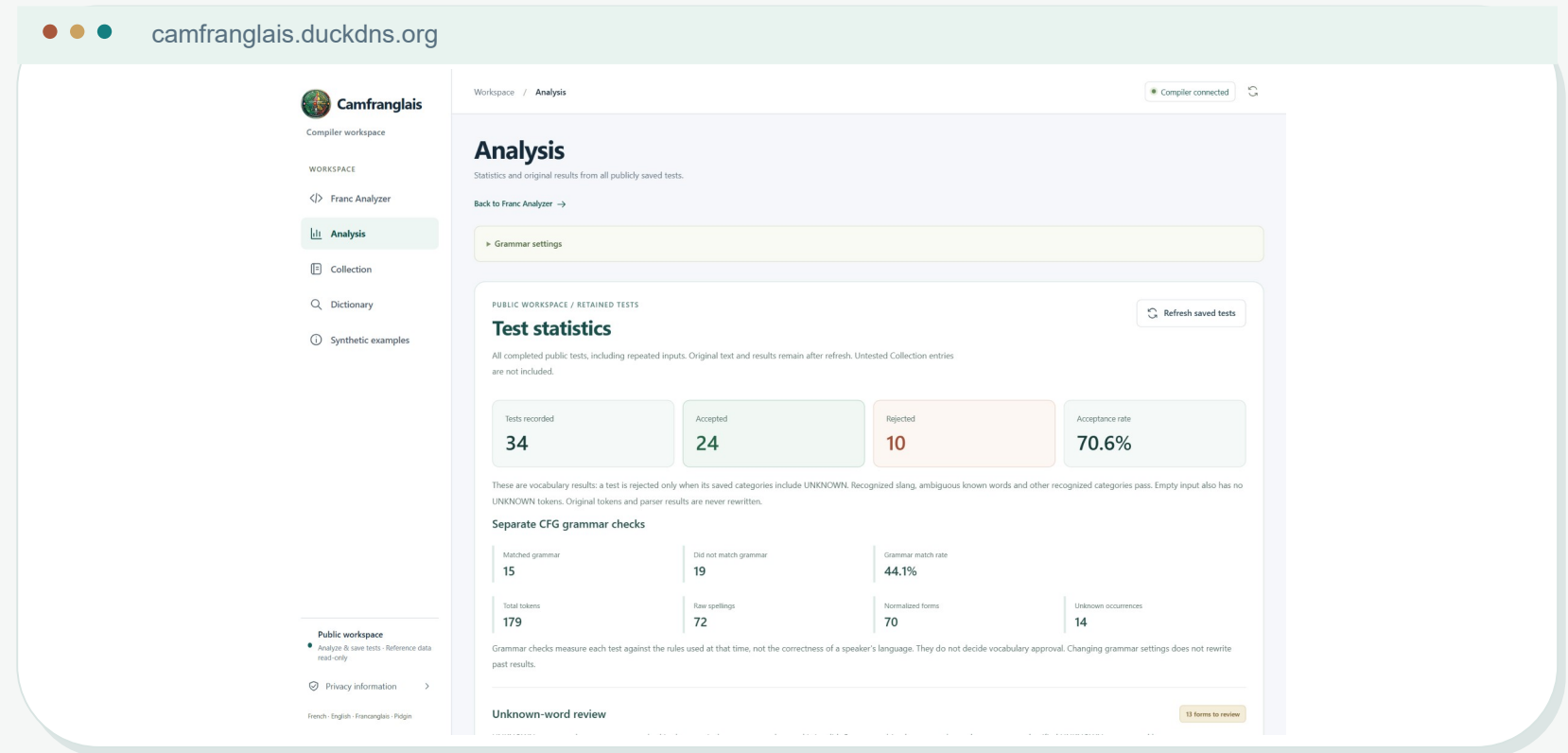
View retained-test totals and verdicts.

Individual result

Inspect the original tokens and trace.

Rule visibility

Open the read-only grammar settings.



REAL DEPLOYED APP / 27 SEP 2026 / crop only where noted

IMPLEMENTATION / EVIDENCE

frontend/src/Analysis.tsx
frontend/src/CourseworkResults.tsx
CS4110 / SETA / 27 SEP 2026

frontend/src/TestStatistics.tsx

Count saved tests, not collected records

A dated history snapshot is different from a controlled study.

34

Retained tests in the screenshot

179

Token occurrences in that history

12

Statements in the fixed study

81

Token occurrences in the fixed study

IMPLEMENTATION / EVIDENCE

frontend/src/TestStatistics.tsx
docs/evidence/final-analysis-20260927.json
CS4110 / SETA / 27 SEP 2026

backend/analyzer_history.py

Vocabulary approval asks one question

Does the saved token stream contain UNKNOWN?

01

ZERO UNKNOWN

Vocabulary ACCEPT.
Slang and AMBIGUOUS are not
automatically rejected.

02

ONE OR MORE UNKNOWN

Vocabulary REJECT.
Keep the unresolved forms visible for
review.

IMPLEMENTATION / EVIDENCE

frontend/src/VocabularyVerdict.tsx

backend/analyzer_models.py

Known words may still fail the CFG

Constructed control: je je suis

01

LEXICAL VIEW

All three words have recognized categories.
Vocabulary approval passes.

02

SYNTACTIC VIEW

The category order is unsupported.
The saved grammar rejects it.

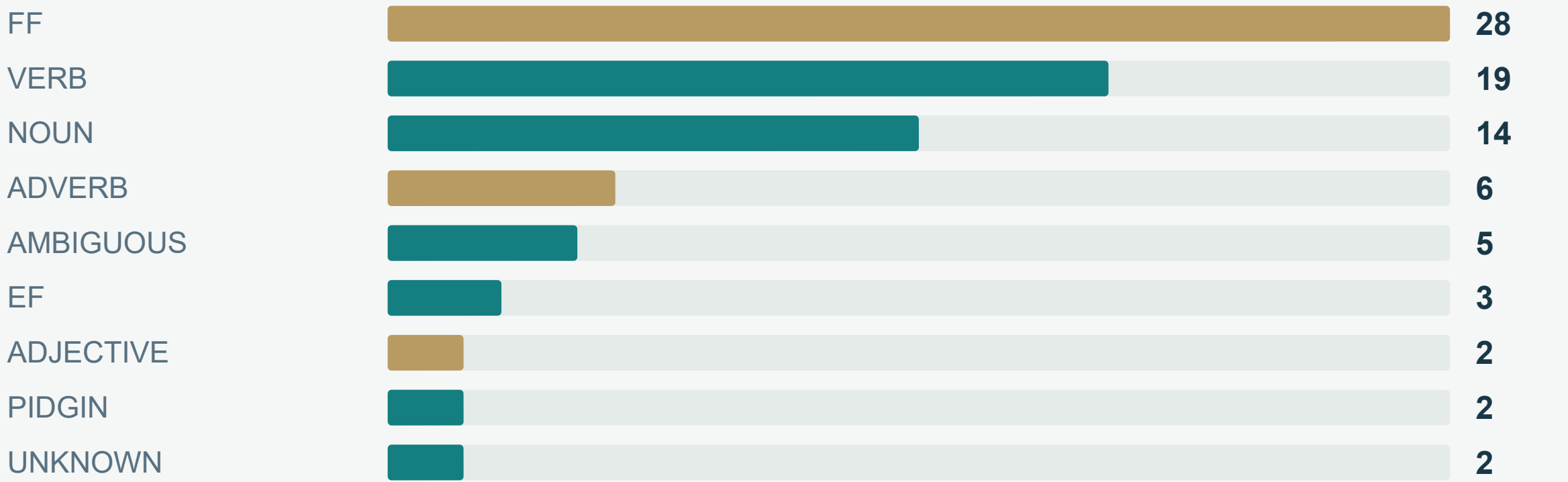
IMPLEMENTATION / EVIDENCE

compiler/parser/yaounde.py
docs/evidence/final-analysis-20260927.json
CS4110 / SET A / 27 SEP 2026

compiler/parser/predictive.py

Nine categories occur in the study

Counts are generated from all 81 token occurrences.



CONTROLLED STUDY / FF = French function word; EF = English function word

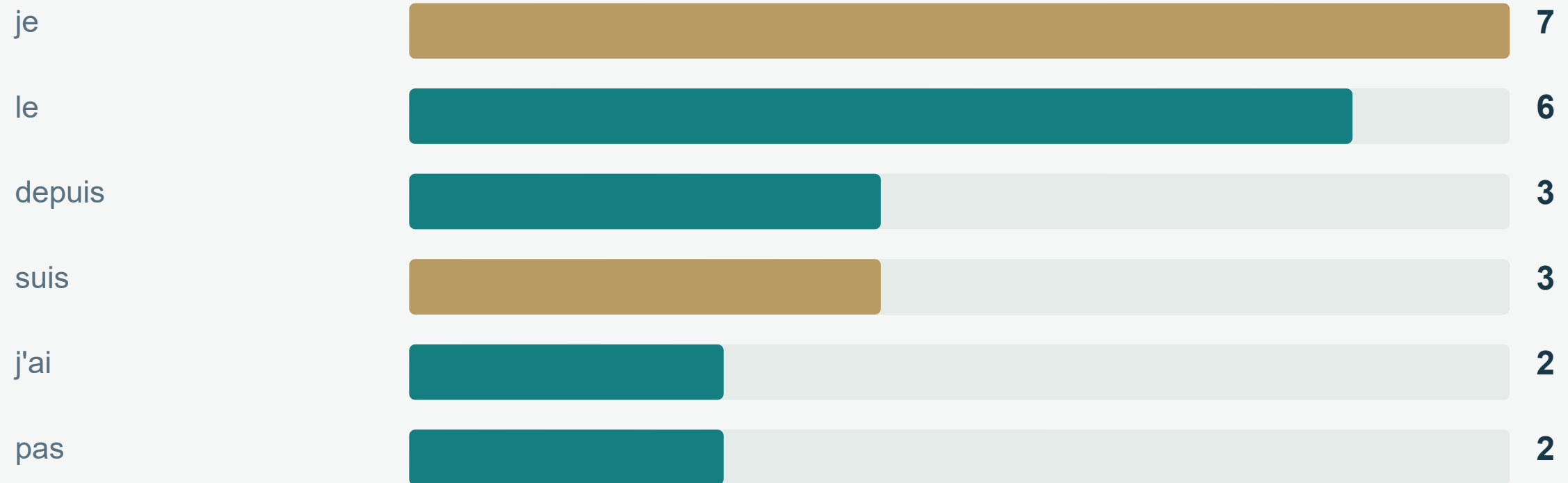
IMPLEMENTATION / EVIDENCE

docs/evidence/final-analysis-20260927.json

tools/build_report_evidence.py

Frequency starts with actual tokens

Most frequent forms in the controlled twelve-statement study.



CONTROLLED STUDY / FF = French function word; EF = English function word

IMPLEMENTATION / EVIDENCE

docs/evidence/final-analysis-20260927.json

tools/build_report_evidence.py

Inspect raw text and categories

The selected-test panel makes individual decisions visible.

Original input

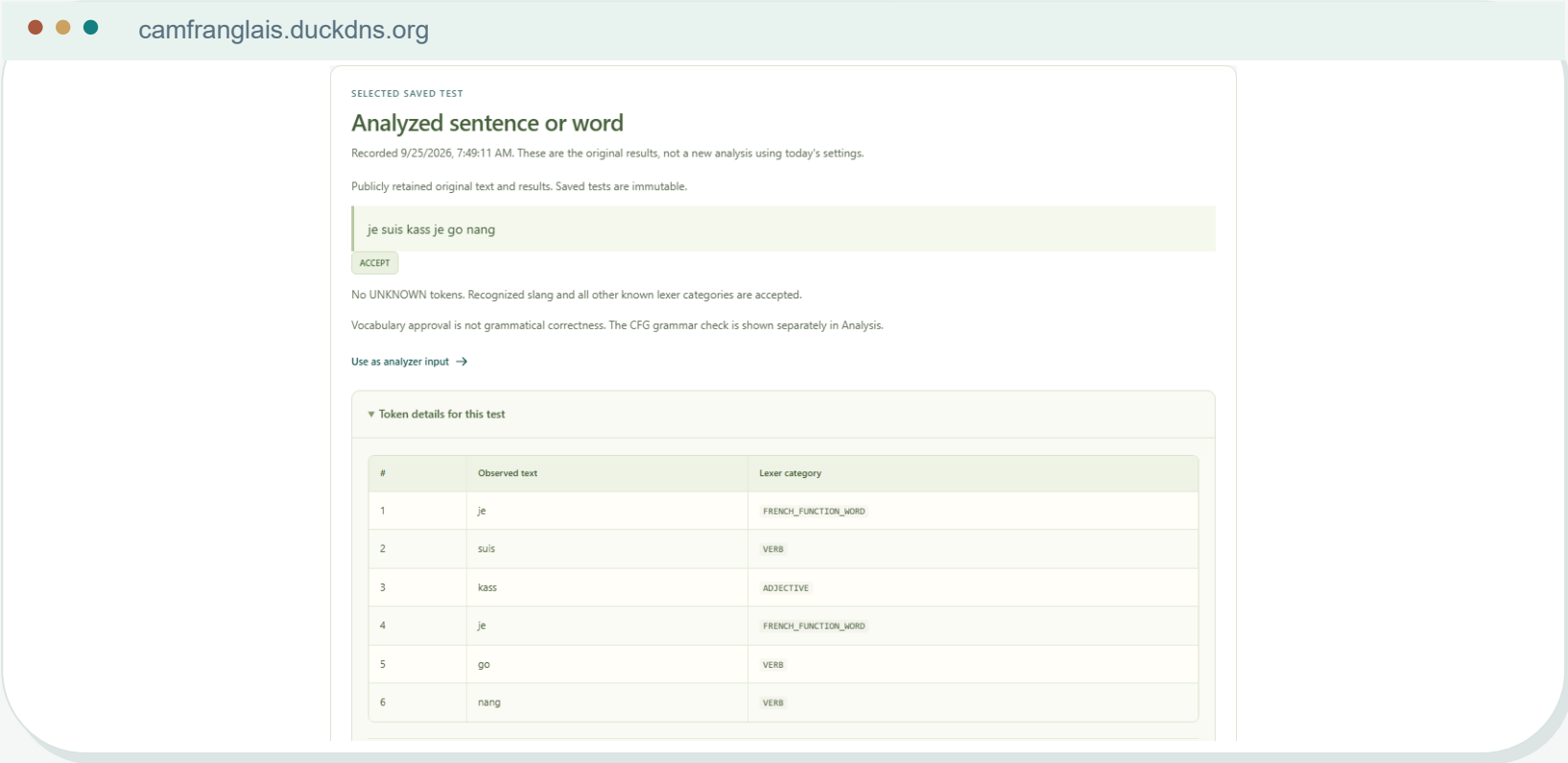
Review exactly what was recorded.

Token detail

Compare each spelling with its chosen label.

Historical result

This is a saved snapshot, not a new computation.



REAL DEPLOYED APP / 27 SEP 2026 / crop only where noted

IMPLEMENTATION / EVIDENCE

frontend/src/CourseworkResults.tsx
backend/analyzer_models.py
CS4110 / SETA / 27 SEP 2026

frontend/src/VocabularyVerdict.tsx

A trace is a sequence of decisions

Each row records the stack and remaining input before its action.

Stack (top at right)	Remaining input	Action
[\$, Ut]	FF VERB ADJECTIVE FF VERB VERB \$	Apply Ut -> CI Uf
[\$, Uf, CI]	FF VERB ADJECTIVE FF VERB VERB \$	Apply CI -> FF St
[\$, Uf, St, FF]	FF VERB ADJECTIVE FF VERB VERB \$	Match FF
[\$, Uf, St]	VERB ADJECTIVE FF VERB VERB \$	Apply St -> VERB Pt

Stack state / remaining categories / selected action.

Ut = Utterance / Uf = ending helper / CI = Clause / St = SubjectTail / FF = French function word

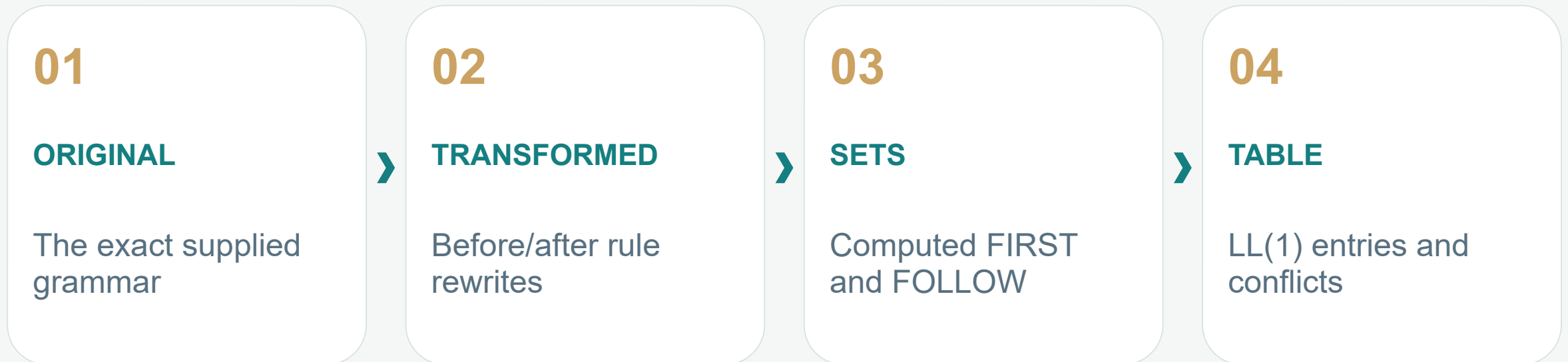
IMPLEMENTATION / EVIDENCE

compiler/parser/predictive.py

frontend/src/CourseworkResults.tsx

Grammar panels reveal the calculation

The UI exposes intermediate results instead of only a green badge.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

frontend/src/CourseworkResults.tsx
compiler/parser/analysis.py
CS4110 / SET A / 27 SEP 2026

compiler/parser/service.py

Analysis: rendering versus aggregation

Different files own the view, the evidence and the totals.

frontend/src

Analysis.tsx

Page wrapper, selected test and loading states.

frontend/src

TestStatistics.tsx

History metrics, frequencies and pagination.

frontend/src

CourseworkResults.tsx

Token tables, grammar tables and parse traces.

backend

analyzer_history.py

Aggregate saved snapshots and page summaries.

IMPLEMENTATION / EVIDENCE

frontend/src/Analysis.tsx
frontend/src/CourseworkResults.tsx
CS4110 / SETA / 27 SEP 2026

frontend/src/TestStatistics.tsx
backend/analyzer_history.py

Collection preserves supplied evidence

A read-only browser for statements, annotations and context.

Browse

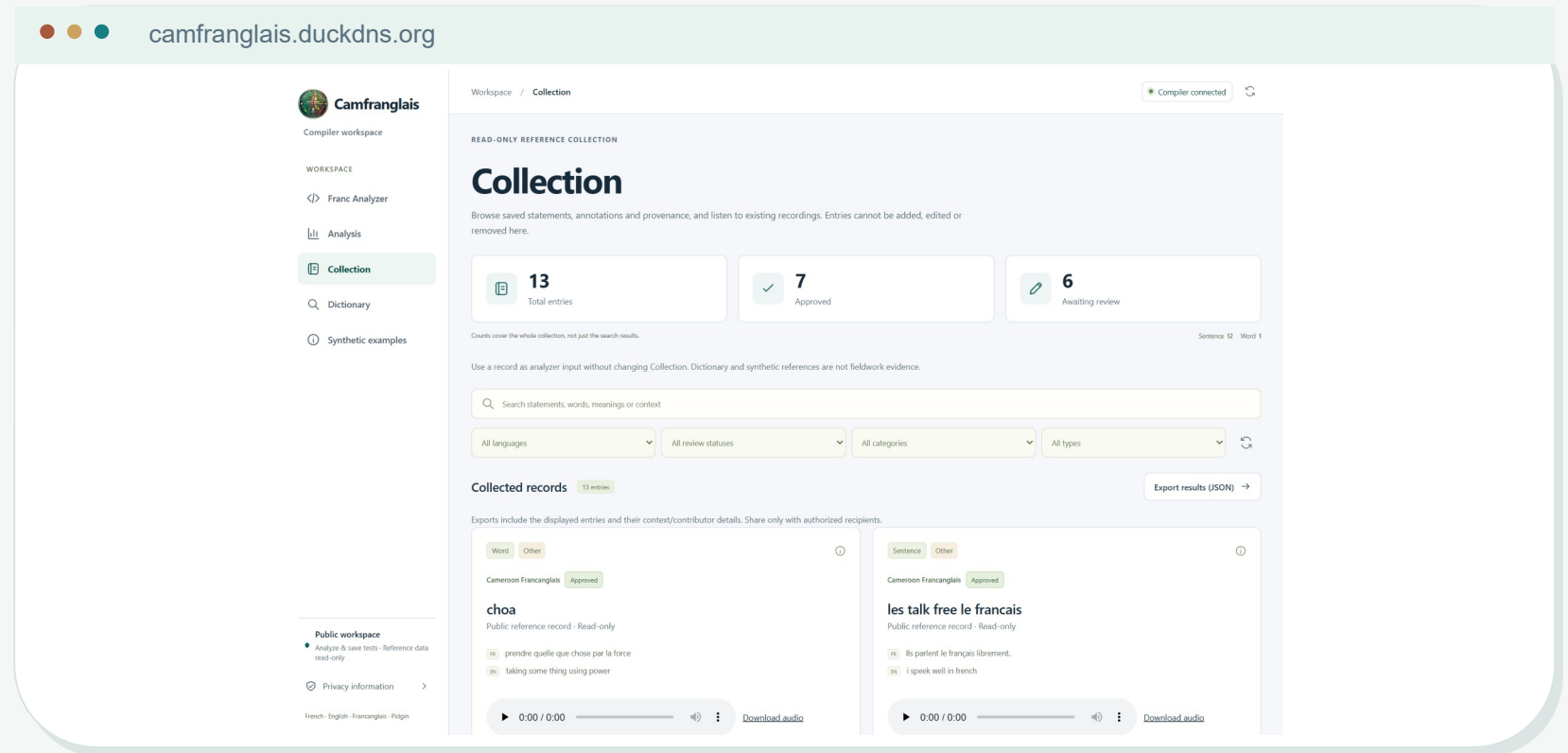
Read the existing records.

Inspect

Open meanings, notes and available audio.

Reuse carefully

Copy a record into an analyzer draft.



REAL DEPLOYED APP / 27 SEP 2026 / crop only where noted

IMPLEMENTATION / EVIDENCE

frontend/src/Collection.tsx

backend/collection.py

What one Collection record means

Keep content, interpretation and provenance separate.

RAW EXPRESSION

The saved statement or word.

SUPPLIED GLOSSES

Meanings provided with the record.

CONTEXT

Location, review state and notes where present.

EXISTING AUDIO

Playback or download when a stored file is available.

IMPLEMENTATION / EVIDENCE

frontend/src/Collection.tsx
backend/audio_api.py
CS4110 / SETA / 27 SEP 2026

backend/schemas.py

Search and filter without rewriting

Read-side tools help visitors find relevant records.

Search and filters

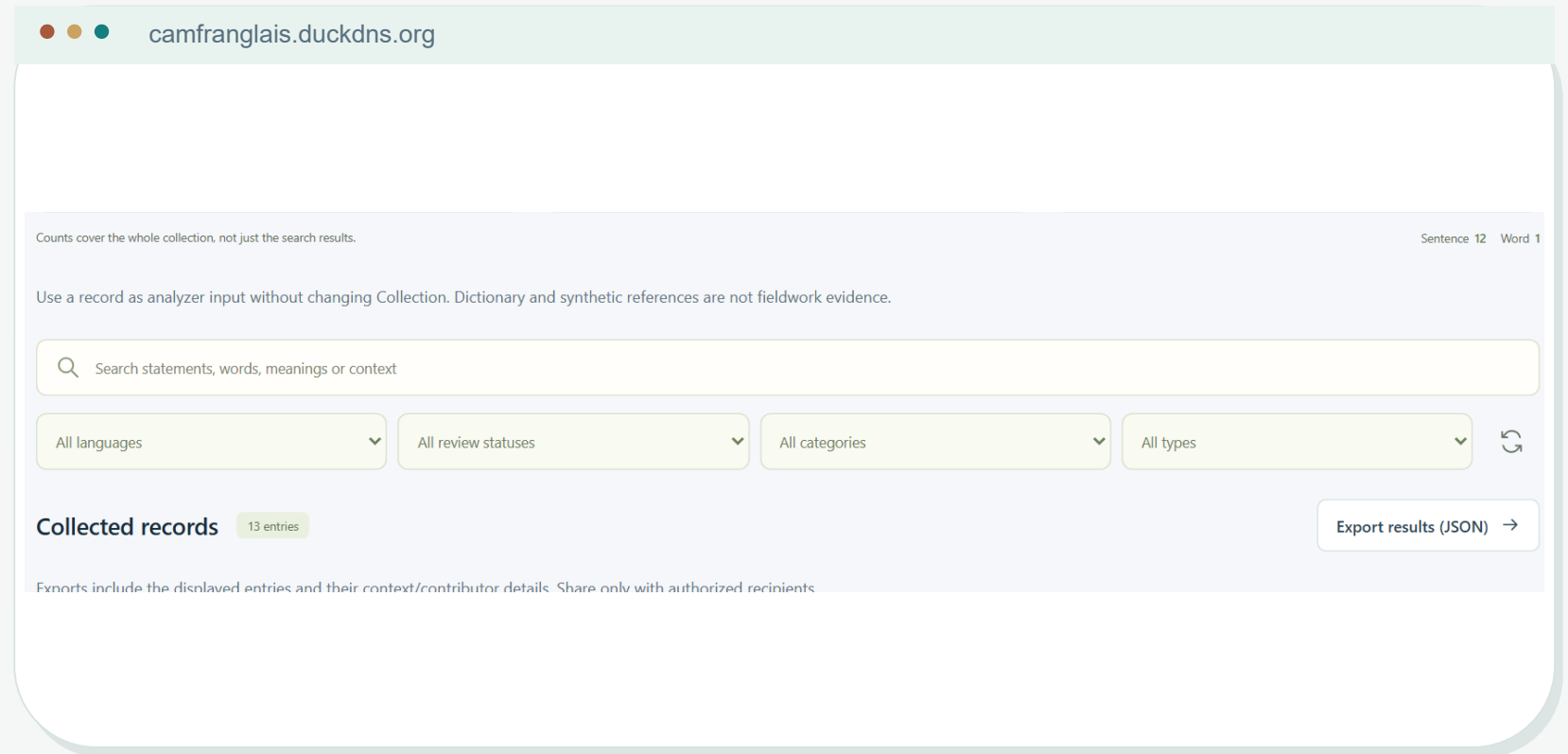
Filter language, review, category and type.

JSON export

Download the filtered public result set.

No mutation

Searching or exporting does not change a record.



REAL DEPLOYED APP / 27 SEP 2026 / crop only where noted

IMPLEMENTATION / EVIDENCE

frontend/src/Collection.tsx

backend/collection.py

Twelve statements; provenance still qualified

Software can measure the corpus without certifying its origin.

12

Exact supplied statements

81

Token occurrences

55

Case-folded forms

UNCONFIRMED

Original fieldwork provenance

IMPLEMENTATION / EVIDENCE

docs/evidence/final-corpus-20260927.json

docs/evidence/final-analysis-20260927.json

Dictionary is a reference library

Search vocabulary, supplied labels and source meanings.

938 entries

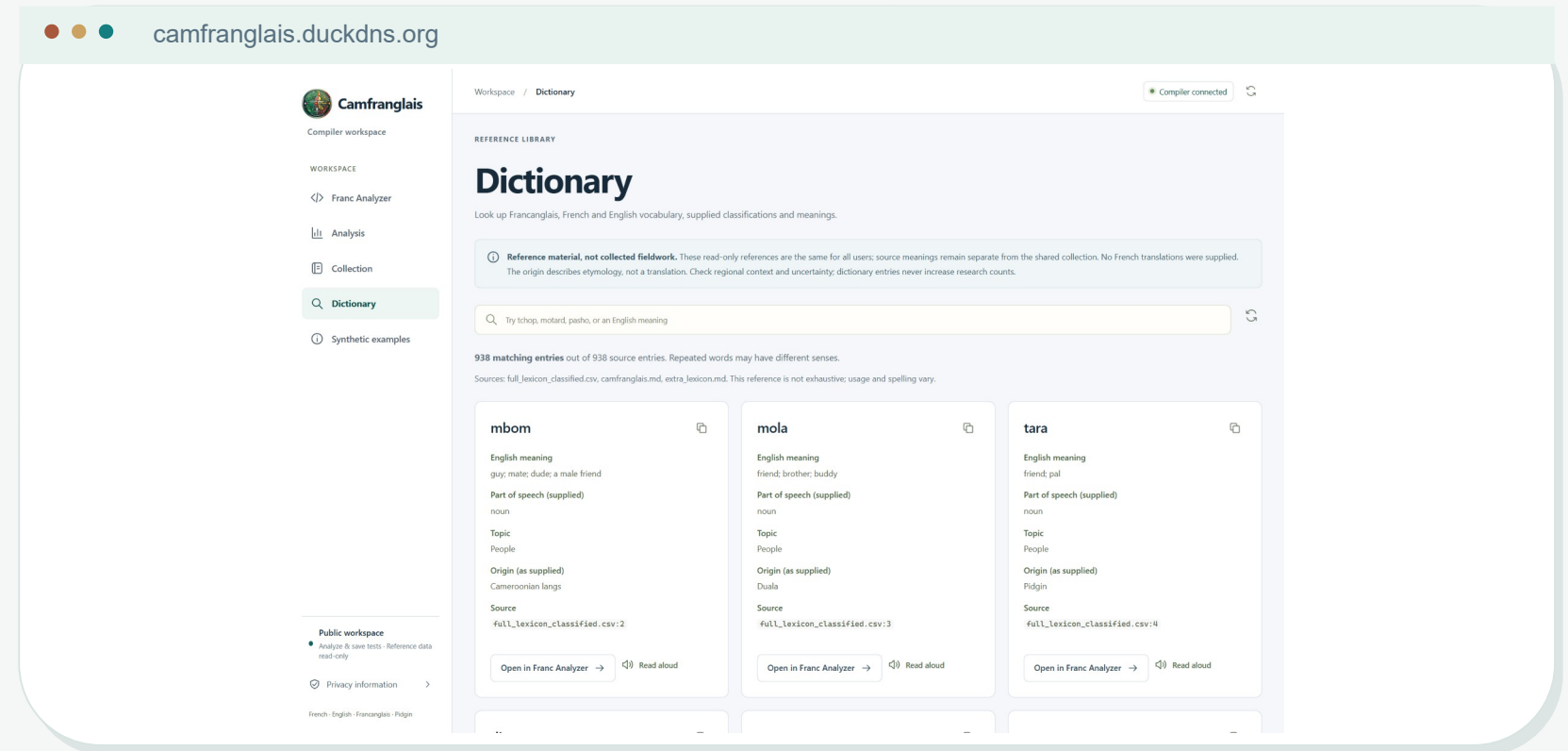
Distinct reference entries, including retained senses.

Source-labelled

See where a meaning or category came from.

Not fieldwork

Reference entries never increase research counts.



REAL DEPLOYED APP / 27 SEP 2026 / crop only where noted

IMPLEMENTATION / EVIDENCE

frontend/src/Dictionary.tsx
dictionary/full_lexicon_classified.csv
CS4110 / SETA / 27 SEP 2026

backend/dictionary.py

Search kass and inspect its source

A concrete lookup connects the interface to the reference data.

Meaning

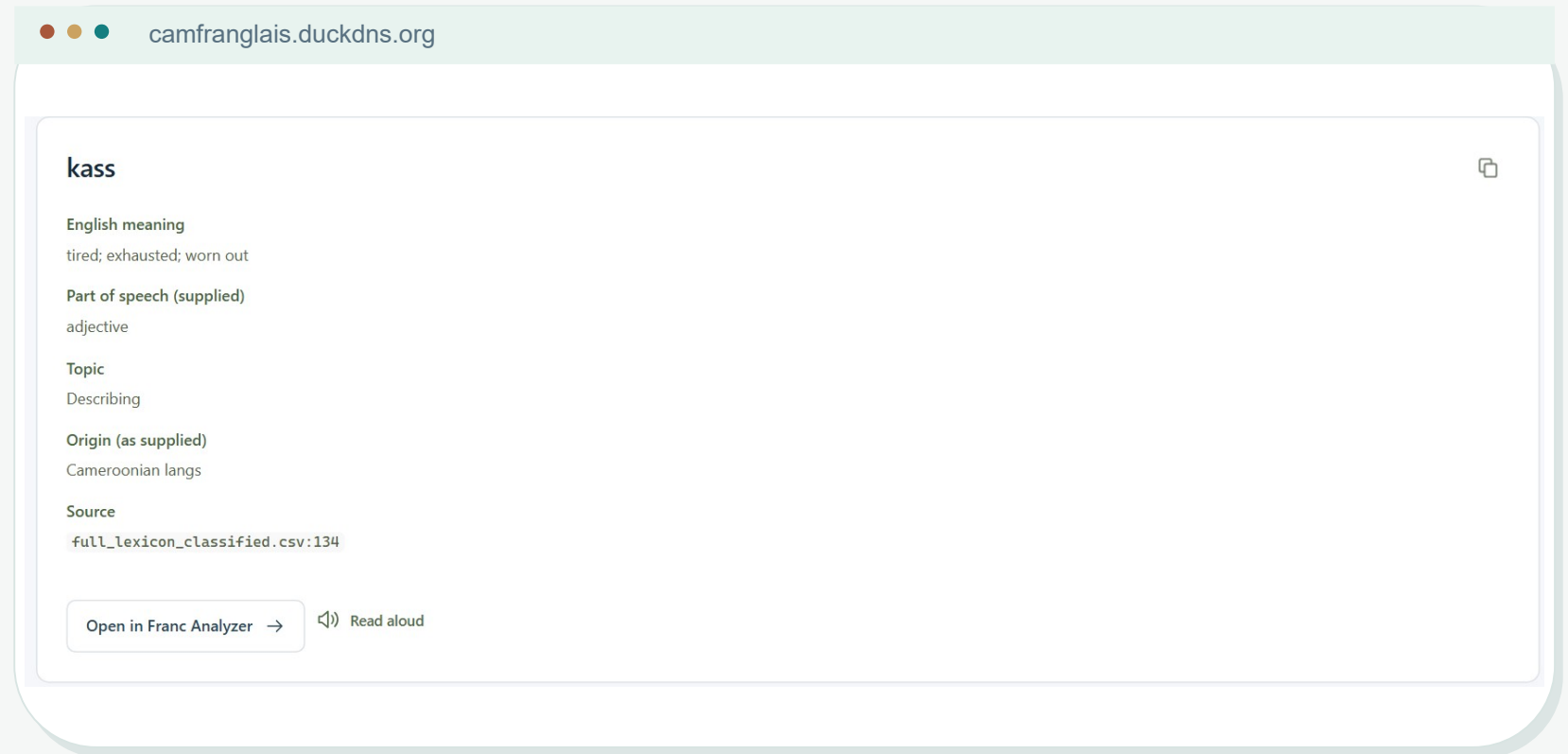
tired; exhausted; worn out

Supplied category

adjective

Traceable source

full_lexicon_classified.csv,
line 134



REAL DEPLOYED APP / 27 SEP 2026 / crop only where noted

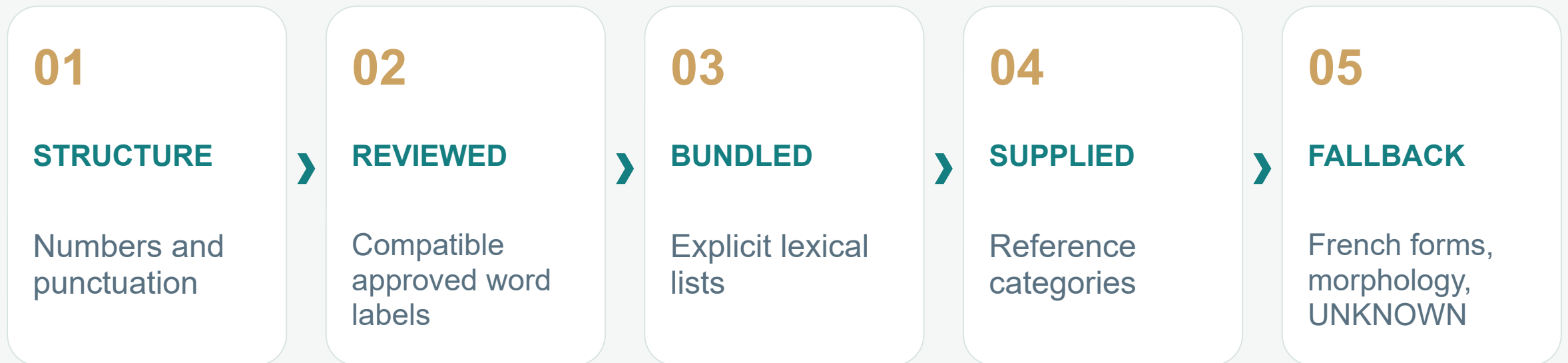
IMPLEMENTATION / EVIDENCE

frontend/src/Dictionary.tsx
dictionary/full_lexicon_classified.csv
CS4110 / SETA / 27 SEP 2026

backend/dictionary.py

Reference lookup does not overrule everything

The classifier uses a defined priority order.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py
compiler/lexer/learned.py
CS4110 / SETA / 27 SEP 2026

compiler/lexer/reference.py

Synthetic means constructed, not collected

Practice examples are useful precisely because they are labelled honestly.

26 examples

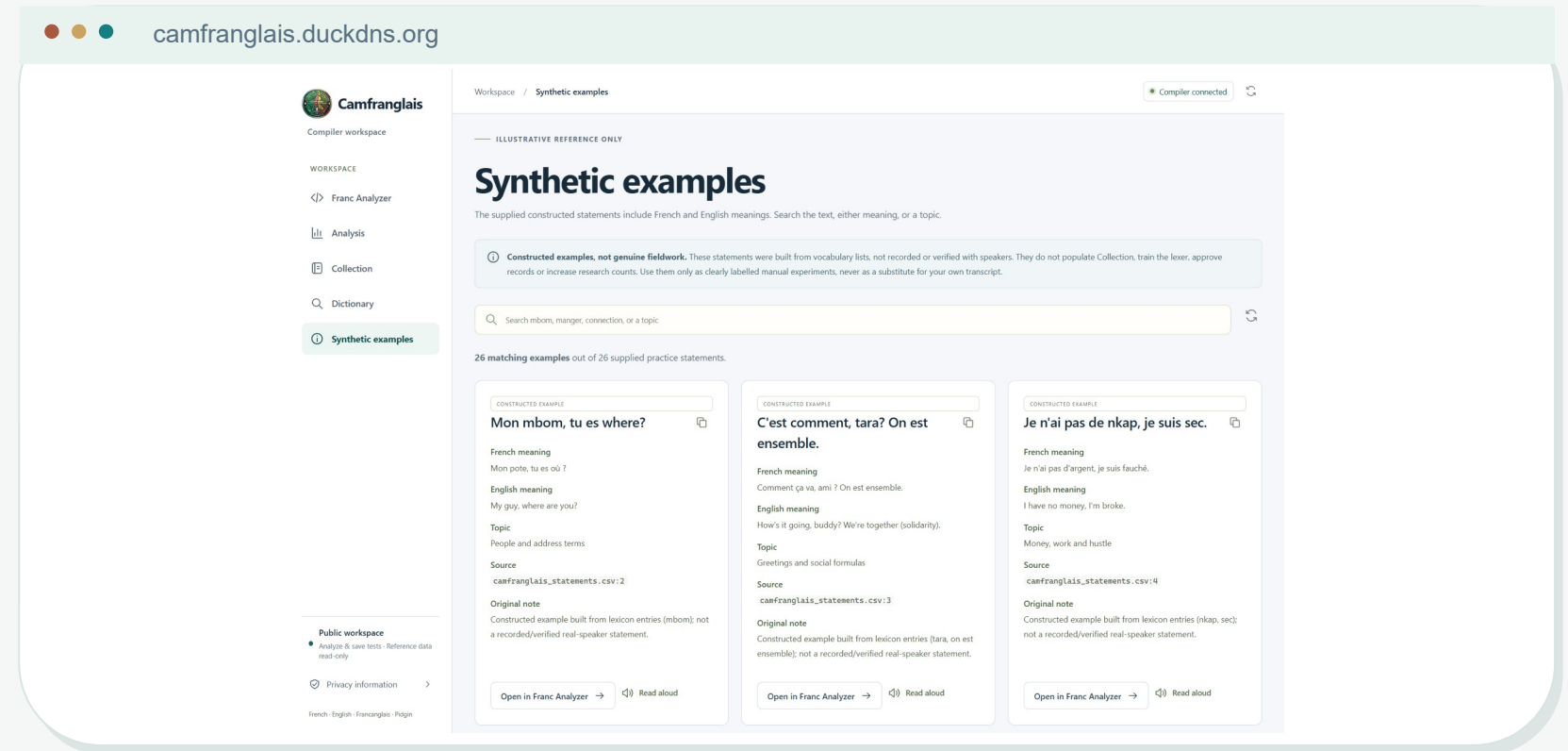
A separate practice library.

Search and browse

Inspect topics, meanings and source labels.

Draft handoff

Try a chosen example without altering Collection.



REAL DEPLOYED APP / 27 SEP 2026 / crop only where noted

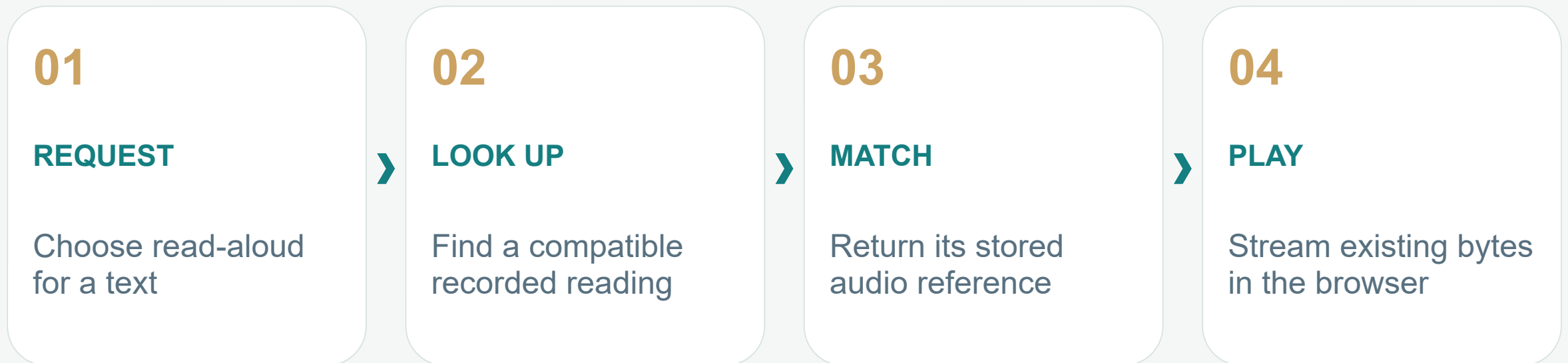
IMPLEMENTATION / EVIDENCE

frontend/src/Examples.tsx

backend/examples.py

Read-aloud reuses an existing recording

There is no generated speech in the public workflow.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

frontend/src/RecordedReadings.tsx
backend/readings_api.py
CS4110 / SETA / 27 SEP 2026

frontend/src/voice.ts

Library features: source ownership

Three libraries share patterns, not the same evidence role.

frontend/src

Collection.tsx

Collection browser and record viewer.

frontend/src

Dictionary.tsx

Reference vocabulary interface.

frontend/src

Examples.tsx

Constructed practice interface.

frontend/src

useReferenceSearch.ts

Debounced reference search and paging.

backend

dictionary.py

Reference assembly and lookup.

backend

examples.py

Constructed example loading and lookup.

IMPLEMENTATION / EVIDENCE

frontend/src/Collection.tsx

backend/dictionary.py

CS4110 / SETA / 27 SEP 2026

frontend/src/useReferenceSearch.ts

backend/examples.py

Three working layers, one product

The browser presents. The server coordinates. The compiler decides.

REACT

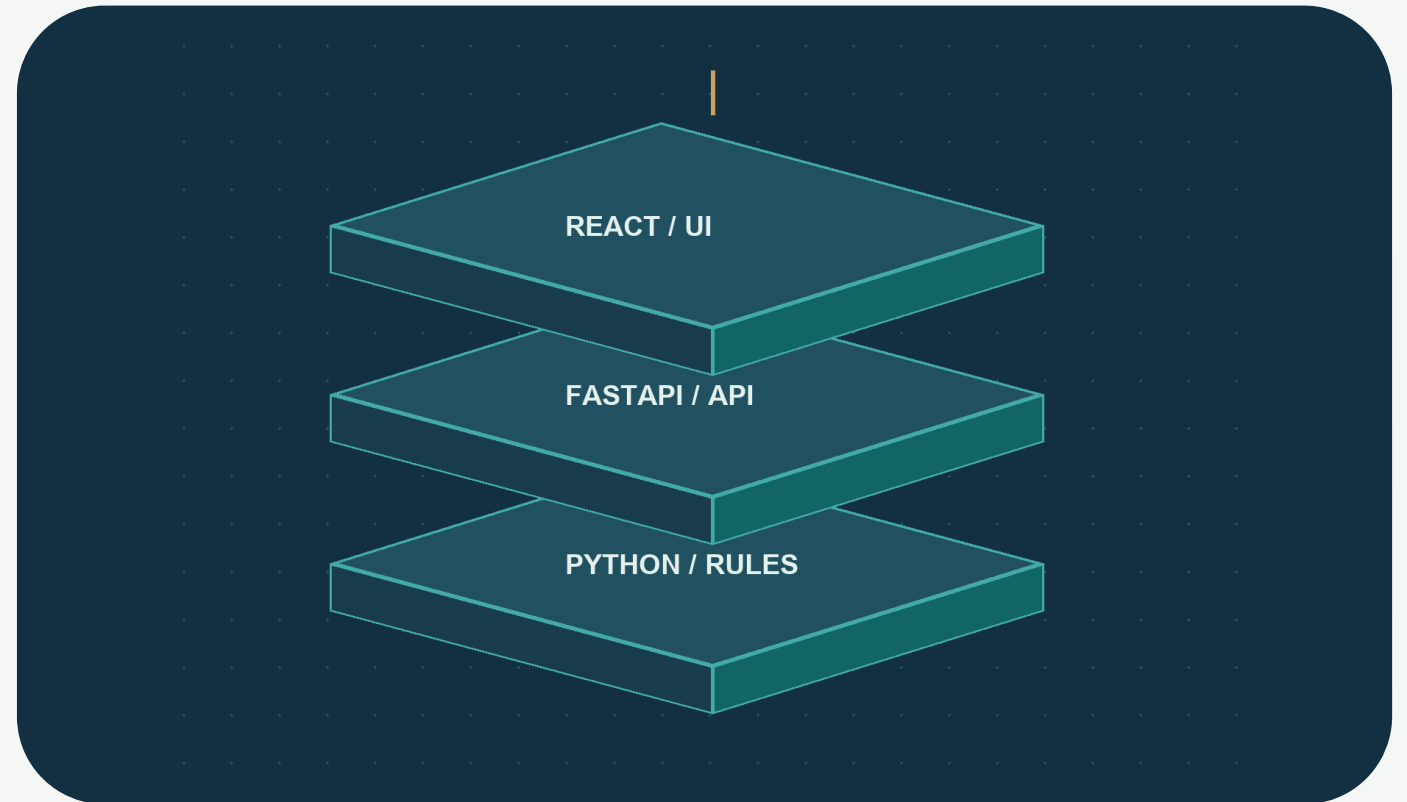
Pages, inputs, feedback and evidence panels.

FASTAPI

Validation, public policy, endpoints and storage.

PYTHON COMPILER

Regex lexing, grammar analysis and predictive parsing.



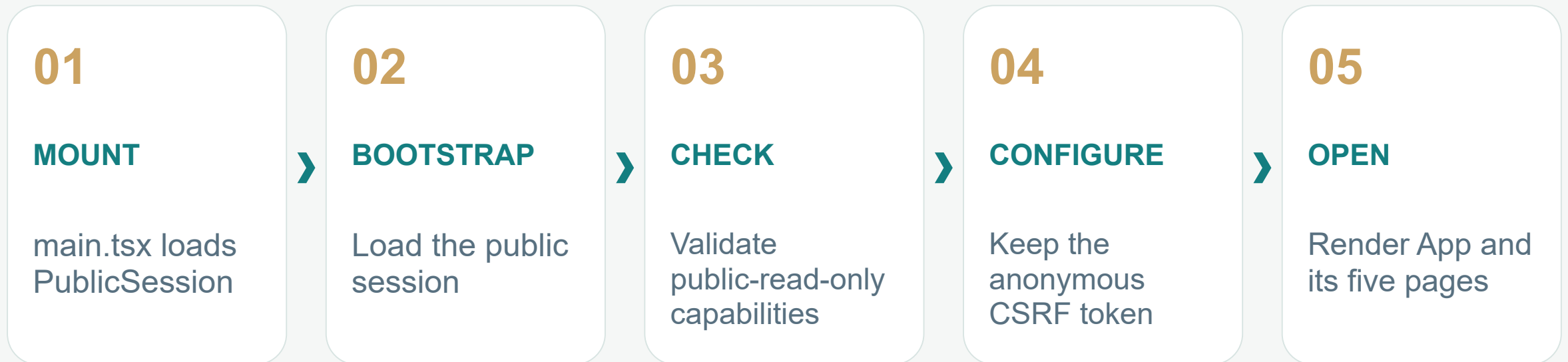
IMPLEMENTATION / EVIDENCE

frontend/src/App.tsx
compiler/parser/service.py
CS4110 / SET A / 27 SEP 2026

backend/main.py

Boot without a login

Public-session preparation replaces account authentication.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

frontend/src/main.tsx
frontend/src/api.ts
CS4110 / SETA / 27 SEP 2026

frontend/src/PublicSession.tsx
backend/public_access.py

Hash routes select the five pages

Navigation stays inside the same application shell.

URL fragment	Visible page	Principal file
#compiler	Franc Analyzer	FrancAnalyzer.tsx
#analysis	Analysis	Analysis.tsx
#collection	Collection	Collection.tsx
#dictionary	Dictionary	Dictionary.tsx
#examples	Synthetic examples	Examples.tsx

IMPLEMENTATION / EVIDENCE

frontend/src/App.tsx

Waiting state and transport are separate

Two shared helpers keep page code focused.

01

useRequest.ts

Owens pending and error state.
Prevents overlapping runs.
Supports cancel and clearError.

02

api.ts

Sends JSON and CSRF headers.
Normalizes errors and timeouts.
Guards against stale sessions.

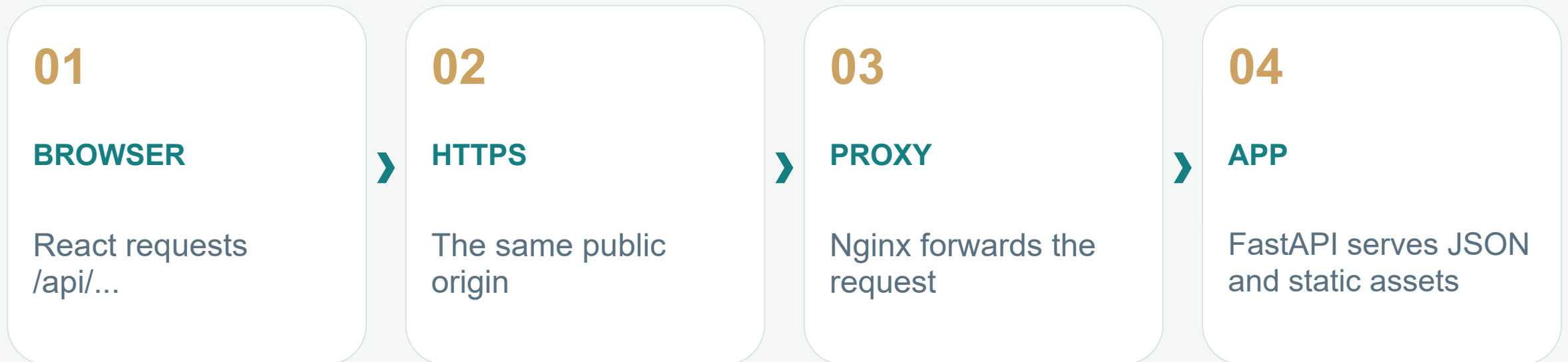
IMPLEMENTATION / EVIDENCE

frontend/src/useRequest.ts

frontend/src/api.ts

One origin connects the UI and API

Production serves the compiled frontend and API together.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

backend/web.py
tools/run_app.py
CS4110 / SETA / 27 SEP 2026

backend/main.py
README.md

Public API, grouped by task

The route reveals which part of the app is involved.

Task	Endpoint	Owner
Session	GET /api/public/session	public_access.py
Record a test	POST /api/analyzer/tests	analyzer_api.py
Test summaries	GET /api/analyzer/tests	analyzer_api.py
Collection	GET /api/dataset	main.py + collection.py
Dictionary	GET /api/dictionary	main.py + dictionary.py
Recorded lookup	POST /api/readings/lookup	readings_api.py

IMPLEMENTATION / EVIDENCE

backend/main.py
backend/public_access.py
CS4110 / SET A / 27 SEP 2026

backend/analyzer_api.py
backend/readings_api.py

Bootstrap advertises the public contract

The frontend verifies capabilities before opening the workspace.

SOURCE WALKTHROUGH / ABBREVIATED WHERE MARKED

```
{
  "access_mode": "public_read_only",
  "capabilities": {
    "analyze": true,
    "save_tests": true,
    "edit_collection": false,
    "edit_grammar": false,
    "edit_recordings": false
  }
}
```

Allowed

Analyze and retain tests.

Protected

Reference data, grammar and recordings.

Anonymous

No account or login is required.

IMPLEMENTATION / EVIDENCE

backend/public_access.py

frontend/src/PublicSession.tsx

Errors tell the user what happened

A failed request is not a success-shaped empty result.

Status	Meaning in this workflow	Next step
403	Forbidden write or invalid Origin/CSRF	Respect the public boundary
404	Missing record or unavailable private API	Check the requested resource
409	Stale grammar or conflicting retry payload	Refresh or reconcile the request
422	Invalid request fields or bounds	Fix the submitted input
429	Rate limit exceeded	Wait for Retry-After
503	Storage temporarily busy	Retry after the service recovers

IMPLEMENTATION / EVIDENCE

backend/public_access.py
frontend/src/api.ts
CS4110 / SET A / 27 SEP 2026

backend/analyzer_history.py

The same workflow on a narrow screen

Responsive presentation without changing the permission model.

Navigate

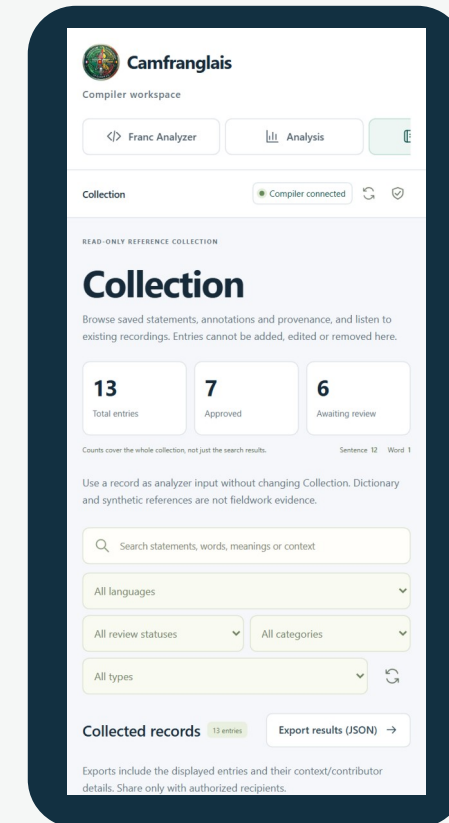
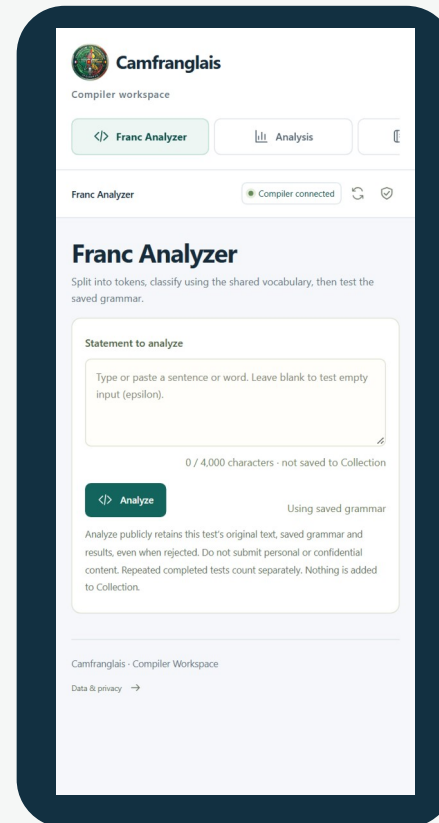
Page controls remain available.

Read

Inputs and cards adapt to width.

Same policy

Reference data remains read-only.



IMPLEMENTATION / EVIDENCE

frontend/src/App.tsx
frontend/src/Collection.tsx
CS4110 / SET A / 27 SEP 2026

frontend/src/FrancAnalyzer.tsx

Frontend file map: find the right layer

Open the smallest responsible surface first.

frontend/src

main.tsx

Mount the public application.

frontend/src

PublicSession.tsx

Validate anonymous capabilities.

frontend/src

App.tsx

Navigation, handoff and shared shell.

frontend/src

api.ts

HTTP transport and CSRF.

frontend/src

useRequest.ts

Request lifecycle in React.

frontend/src

CourseworkResults.tsx

Render compiler evidence.

IMPLEMENTATION / EVIDENCE

frontend/src/main.tsx
frontend/src/App.tsx
CS4110 / SETA / 27 SEP 2026

frontend/src/PublicSession.tsx
frontend/src/api.ts

Words become a typed token stream

The lexer is the first algorithmic stage.

SEGMENT

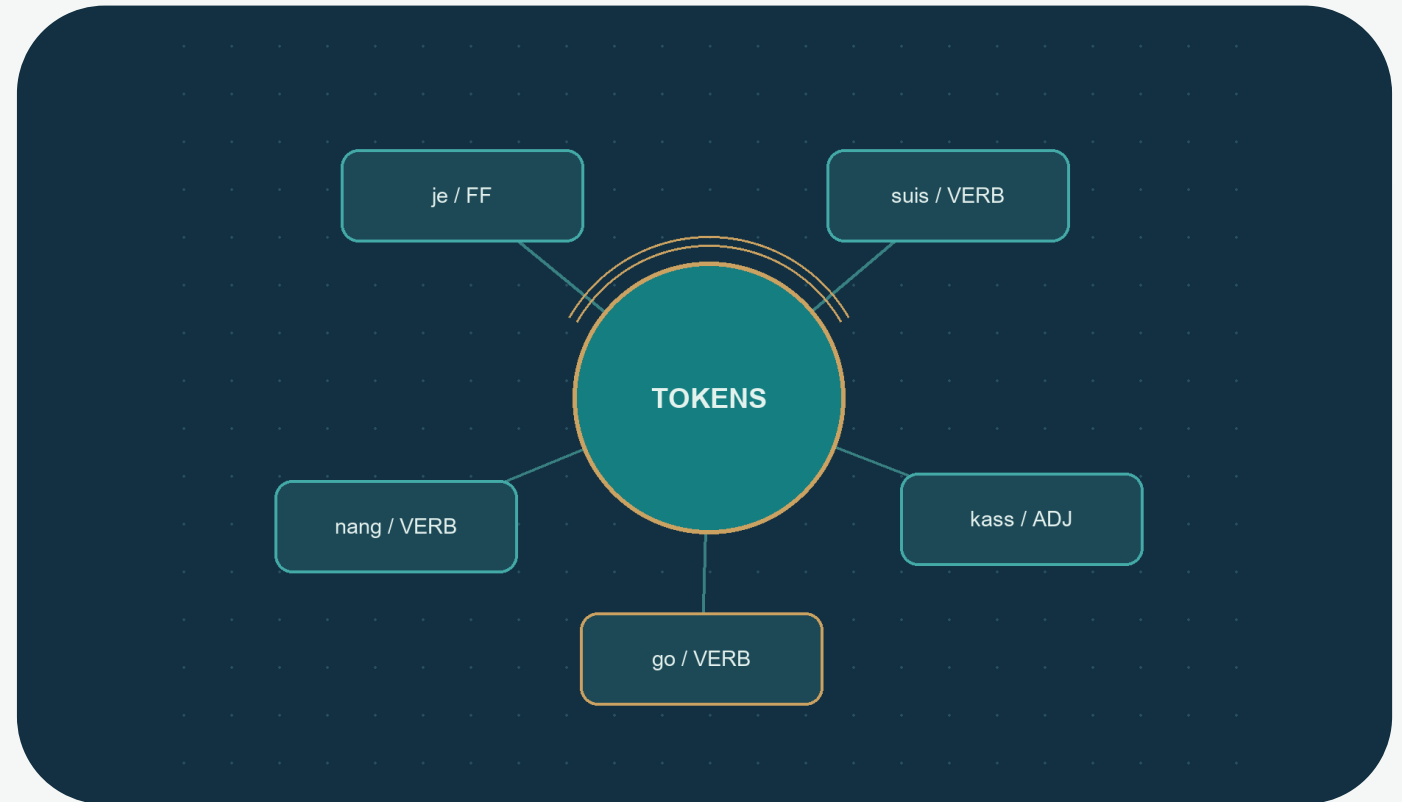
Recognize word forms, numbers and punctuation.

CLASSIFY

Assign a category using an ordered rule set.

ANNOTATE

Add language and phrase clues without deleting tokens.



IMPLEMENTATION / EVIDENCE

`compiler/lexer/tokenizer.py`

`compiler/lexer/lexicon.py`

Read the real segmentation branches

TOKEN_SPLIT_RE preserves unsupported non-whitespace input.

SOURCE WALKTHROUGH / ABBREVIATED WHERE MARKED

```
WORD:    letters + internal joiners
NUMBER:  \d+(?:[.,]\d+)?
PUNCT:   [.,!?:;"'()]
OTHER:   \S
```

```
TOKEN_SPLIT_RE.findall(text)
```

Ordered alternatives

Words, then numbers, punctuation and fallback.

Whitespace

Skipped between matches.

Fallback

An unsupported character survives for classification.

IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py

Internal joiners belong to the word

Small boundary cases make the scanner behavior concrete.

Raw input	Actual tokenization	Reason
j'ai	[j'ai]	Internal apostrophe with continuation
go-slow	[go-slow]	Internal hyphen with continuation
a-	[a, -]	Dangling hyphen cannot finish a word
a--b	[a, -, -, b]	Repeated joiners break the word path

IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py

tools/tests/test_report_evidence.py

Numbers are not signed literals

The decimal branch requires digits on both sides of a separator.

Raw input	Actual tokens	Interpretation
3.14	[3.14]	One decimal-number token
3.	[3, .]	Integer followed by punctuation
.5	[., 5]	Leading punctuation and integer
-5	[-, 5]	Other character followed by integer

IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py

docs/report-data/lexer-controls.tex

Unicode details change the boundary

Not every apostrophe-like character takes the same transition.

01

U+0027 / U+2018 / U+2019

Pending joiners outside the word class.

Need a following word character.

02

U+02BC

Already belongs to Python's word-character class.

Follows the C edges, even at a boundary.

IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py
tools/tests/test_report_evidence.py
CS4110 / SETA / 27 SEP 2026

compiler/lexer/lexicon.py

Classification is ordered, not voted

The first applicable rule determines the token category.

01 / Structural

NUMBER and PUNCTUATION regex matches.

02 / Reviewed

Compatible approved Word annotations.

03 / Explicit lists

Slang, Pidgin, nouns, verbs and function words.

04 / Supplied reference

One category, or AMBIGUOUS for conflicting labels.

05 / Later fallbacks

Observed French forms, then suffix heuristics.

06 / Unresolved

UNKNOWN if no rule applies.

IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py
compiler/lexer/reference.py
CS4110 / SETA / 27 SEP 2026

compiler/lexer/learned.py

AMBIGUOUS is not UNKNOWN

Preserve uncertainty instead of pretending all labels mean the same thing.

01

AMBIGUOUS

The supplied reference has conflicting categories.
It is a known label, not a wildcard.

02

UNKNOWN

No supported classification matched.
It fails vocabulary approval.

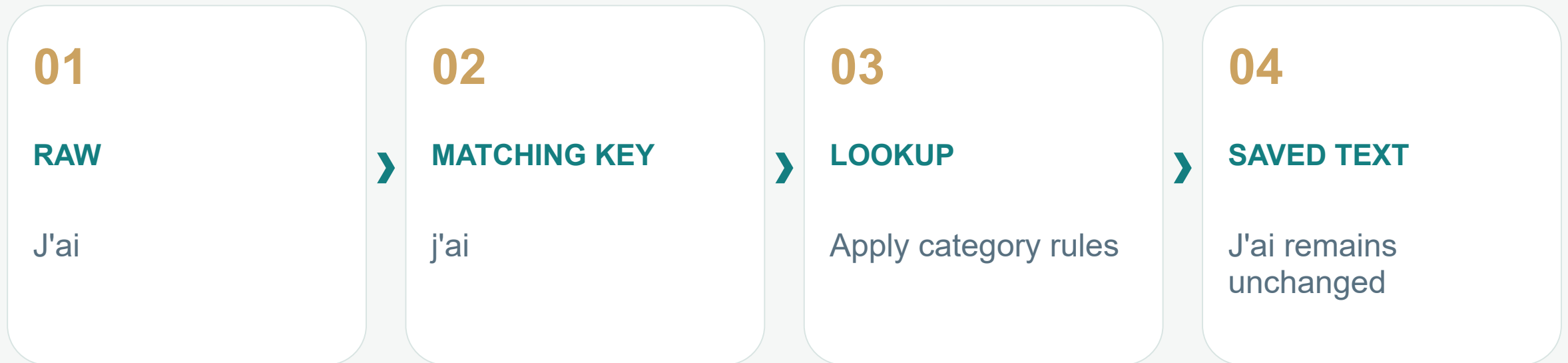
IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py
backend/analyzer_models.py
CS4110 / SET A / 27 SEP 2026

compiler/parser/yaounde.py

Normalize for matching. Preserve for evidence.

A comparison key and the raw text serve different purposes.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

compiler/lexer/vocabulary.py
backend/analyzer_models.py
CS4110 / SETA / 27 SEP 2026

compiler/lexer/tokenizer.py

Language clues and phrases are annotations

Additional evidence must not collapse the token stream.

01

LANGUAGE CLUES

Compare
candidate-language sets;
retain uncertainty.

02

SENTENCE BOUNDARIES

Sentence-ending
punctuation resets the
language run.

03

PHRASE MATCHES

Preserve source-ordered
raw spans as extra
annotations.

IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py
compiler/lexer/reference.py
CS4110 / SETA / 27 SEP 2026

compiler/lexer/languages.py

Lexer files and responsibilities

A small set of modules explains the complete lexical path.

compiler/lexer

tokenizer.py

Segment, classify and annotate.

compiler/lexer

lexicon.py

Terminal categories and explicit rules.

compiler/lexer

reference.py

Read supplied reference categories.

compiler/lexer

learned.py

Build compatible reviewed-word overrides.

compiler/lexer

vocabulary.py

Normalize matching keys and aliases.

compiler/lexer

languages.py

Return candidate-language evidence.

IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py
compiler/lexer/learned.py
CS4110 / SET A / 27 SEP 2026

compiler/lexer/reference.py
compiler/lexer/vocabulary.py

Grammar recognizes category sequences

The parser reads labels, not the meaning of words.

INPUT

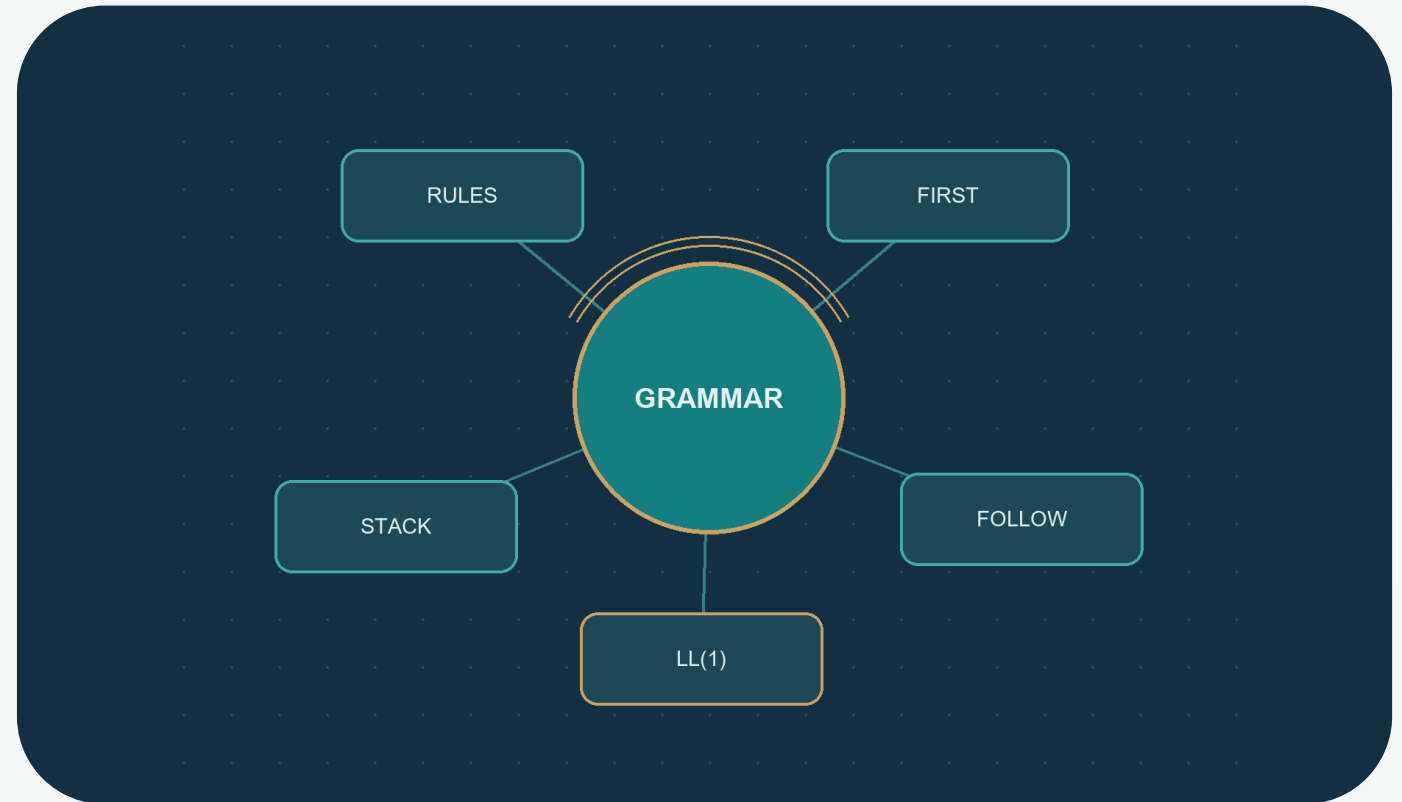
A finite stream of lexer categories.

RULES

A context-free grammar of those categories.

RESULT

Acceptance or an explicit failed transition.



IMPLEMENTATION / EVIDENCE

compiler/parser/grammar.py
compiler/parser/predictive.py
CS4110 / SET A / 27 SEP 2026

compiler/parser/service.py

The saved grammar is corpus-derived

Reusable category structures, not a list of accepted sentences.

SOURCE WALKTHROUGH / ABBREVIATED WHERE MARKED

```
Utterance -> Clause
           | Clause PUNCTUATION

Clause -> FRENCH_FUNCTION_WORD SubjectTail
         | Nominal BareSubjectTail
         | VERB PredicateTail
```

Observed structures

Subject-led, nominal-led and verb-initial clauses.

Limited ending

At most one optional punctuation token.

No forced repair

The two unresolved forms stay rejected.

IMPLEMENTATION / EVIDENCE

compiler/parser/yaounde.py

docs/evidence/final-corpus-20260927.json

Starter grammar and saved grammar differ

Do not confuse a fresh installation with the documented corpus state.

01

TEACHING STARTER

DEFAULT_GRAMMAR in service.py.
Illustrative rules for an empty setup.

02

DOCUMENTED CORPUS

GRAMMAR in yaounde.py.
The group's saved category-level rules.

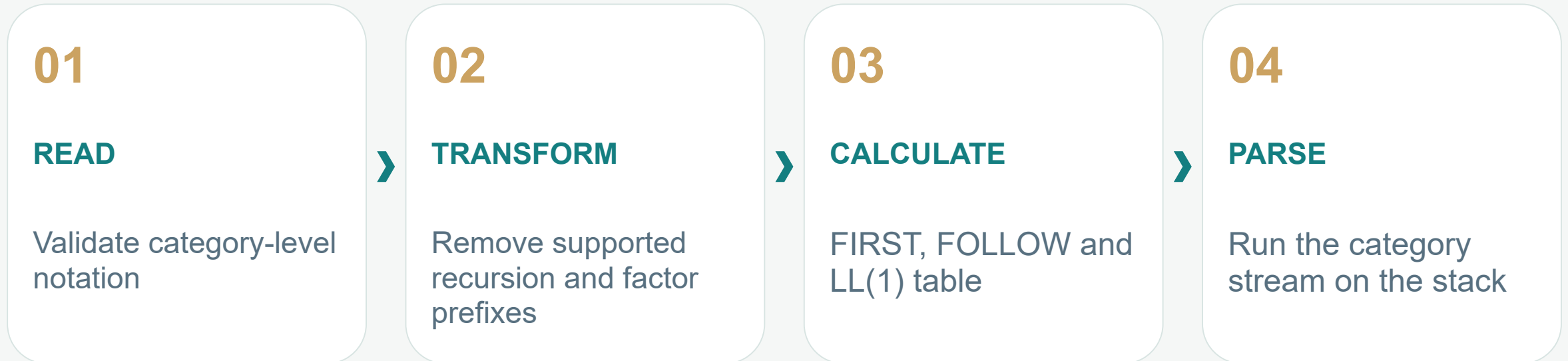
IMPLEMENTATION / EVIDENCE

compiler/parser/service.py
docs/evidence/final-corpus-20260927.json
CS4110 / SETA / 27 SEP 2026

compiler/parser/yaounde.py

Read, transform, calculate, then parse

Each phase produces inspectable output.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

compiler/parser/grammar.py
compiler/parser/service.py
CS4110 / SETA / 27 SEP 2026

compiler/parser/analysis.py
compiler/parser/predictive.py

Remove the actual left recursion

A noun cluster should not recurse before consuming input.

BEFORE / ORIGINAL RULE

```
Nominal -> Nominal NOUN | NOUN
```

AFTER / TRANSFORMED RULES

```
Nominal -> NOUN Nominal_LR1  
Nominal_LR1 -> NOUN Nominal_LR1 | epsilon
```

MEANING PRESERVED

One or more noun categories.

PREDICTIVE SHAPE

Consume a NOUN before repeating.

IMPLEMENTATION / EVIDENCE

compiler/parser/yaounde.py

compiler/parser/analysis.py

Factor the shared Clause prefix

Choose the ending only after the common clause has been parsed.

BEFORE / ORIGINAL RULE

```
Utterance -> Clause | Clause PUNCTUATION
```

AFTER / TRANSFORMED RULES

```
Utterance -> Clause Utterance_LF1  
Utterance_LF1 -> epsilon | PUNCTUATION
```

SHARED PREFIX

Both alternatives begin with Clause.

FACTORED CHOICE

The helper selects no ending or one punctuation token.

IMPLEMENTATION / EVIDENCE

compiler/parser/yaounde.py

compiler/parser/analysis.py

FIRST tells us what can start

For the transformed study grammar:

FIRST(Utterance)

FRENCH_FUNCTION_WORD

A subject-led clause.

NOUN

A nominal-led clause.

VERB

A verb-initial clause.

IMPLEMENTATION / EVIDENCE

compiler/parser/analysis.py

docs/evidence/final-analysis-20260927.json

FOLLOW tells us what can come next

End-of-input matters when a production is nullable.

$$\text{FOLLOW}(\text{Utterance}) = \{ \$ \}$$

START SYMBOL

Its FOLLOW set includes the end marker.

NULLABLE RHS

The LL(1) builder also uses FOLLOW lookaheads.

NO EXTRA TOKEN

\$ is a virtual marker, not a word typed by the visitor.

IMPLEMENTATION / EVIDENCE

compiler/parser/analysis.py

compiler/parser/grammar.py

The study's transformed grammar

Measured structure after the two recorded rewrites.

12

Nonterminals

39

Productions

47

Populated LL(1) cells

0

Table conflicts

IMPLEMENTATION / EVIDENCE

`compiler/parser/analysis.py`

`docs/evidence/final-analysis-20260927.json`

Syntax-analysis files: a map

Locate notation, transformations, orchestration and execution.

compiler/parser

grammar.py

Read and validate grammar notation.

compiler/parser

analysis.py

Transform rules; compute sets and table.

compiler/parser

service.py

Assemble JSON-serializable analysis.

compiler/parser

predictive.py

Execute the bounded LL(1) stack.

compiler/parser

yaounde.py

Corpus grammar, rationale and limits.

IMPLEMENTATION / EVIDENCE

compiler/parser/grammar.py
compiler/parser/service.py
CS4110 / SETA / 27 SEP 2026

compiler/parser/analysis.py
compiler/parser/predictive.py

Lexer DFA: recognizing token forms

An equivalent formal model of the regex, not a runtime DFA class.

Double circles

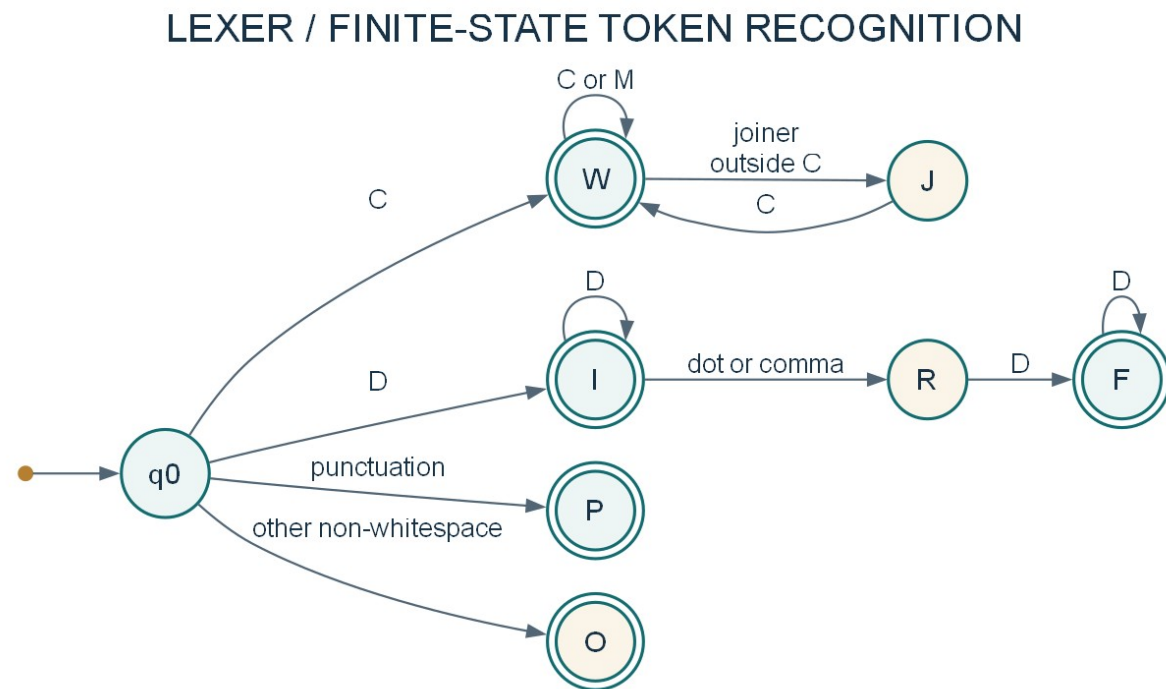
Accept a token form.

J and R

Pending joiner or decimal continuation.

Implicit dead state

Unshown transitions stop this token path.



IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py

docs/diagrams/automaton-lexer.puml

A pending state cannot end a token

Compare 3.14 with 3. using the same automaton.

01

I

Read 3: integer
accepts



02

R

Read .: separator is
pending



03

F

Read 14: fraction
accepts



04

RESCAN

With no following digit,
emit 3 then scan .
again

FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

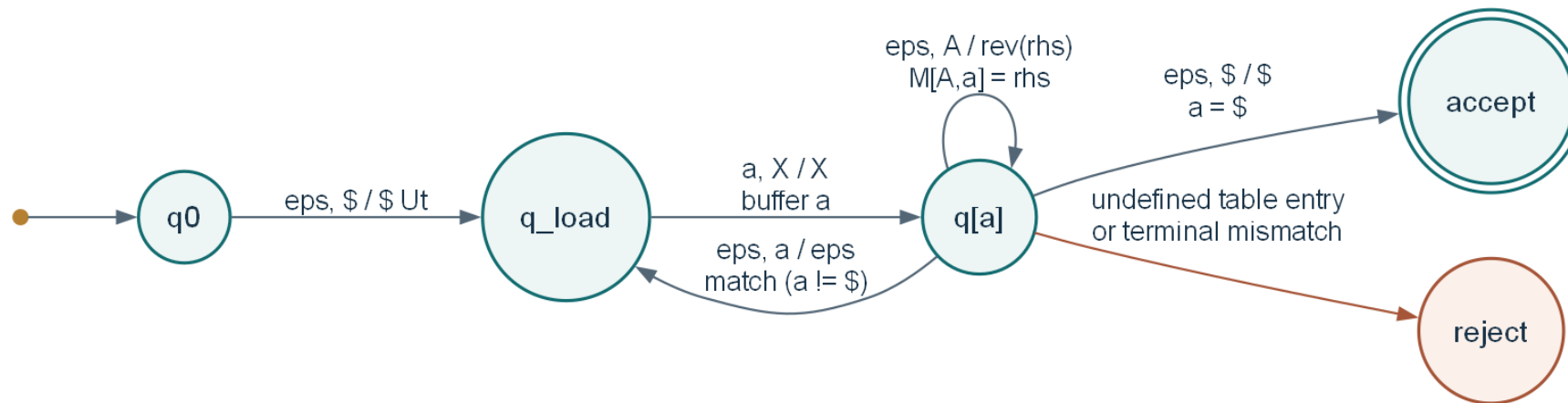
compiler/lexer/tokenizer.py
docs/report-data/lexer-controls.tex
CS4110 / SETA / 27 SEP 2026

docs/diagrams/automaton-lexer.puml

The parser adds a stack

Deterministic pushdown control explains the saved LL(1) recognizer.

PARSER / DETERMINISTIC PUSHDOWN CONTROL



EXPAND

Use the unique table cell.

MATCH

Pop a terminal and advance.

ACCEPT

Bottom marker and input end agree.

IMPLEMENTATION / EVIDENCE

compiler/parser/predictive.py

docs/diagrams/automaton-parser.puml

The top of the stack is on the right

S02 begins with a start symbol above the bottom marker.

BEFORE

\$

Ut

TOP

AFTER

\$

Uf

Cl

TOP

Clause must be handled before the ending helper.

One table entry controls expansion

Nonterminal expansion does not consume a category.

SOURCE WALKTHROUGH / ABBREVIATED WHERE MARKED

```
production = table[top][lookahead]

record("Apply ...")
stack.pop()
stack.extend(reversed(production))
```

Lookahead

Choose a unique $M[A, a]$ entry.

Replacement

Push the RHS in reverse order.

Epsilon

Pop A without consuming input.

IMPLEMENTATION / EVIDENCE

compiler/parser/predictive.py

compiler/parser/analysis.py

A terminal match consumes one category

The stack and logical input advance together.

BEFORE

\$

Uf

St

TOP
FF

AFTER

\$

Uf

St

TOP

The next category is VERB.

IMPLEMENTATION / EVIDENCE

compiler/parser/predictive.py

docs/evidence/final-analysis-20260927.json

Walk the first four S02 actions

Expansion and matching leave different fingerprints in the trace.

Stack (top at right)	Remaining input	Action
[\$, Ut]	FF VERB ADJECTIVE FF VERB VERB \$	Apply Ut -> CI Uf
[\$, Uf, CI]	FF VERB ADJECTIVE FF VERB VERB \$	Apply CI -> FF St
[\$, Uf, St, FF]	FF VERB ADJECTIVE FF VERB VERB \$	Match FF
[\$, Uf, St]	VERB ADJECTIVE FF VERB VERB \$	Apply St -> VERB Pt

Only Match FF advances the remaining input.

Ut = Utterance / Uf = ending helper / CI = Clause / St = SubjectTail / FF = French function word

IMPLEMENTATION / EVIDENCE

Acceptance and rejection are explicit

Both outcomes return evidence rather than a guess.

01

S02 / ACCEPT

6 consumed tokens.
16 recorded trace rows.
Final stack and input both show \$.

02

S09 / REJECT

5 consumed tokens.
14 recorded trace rows.
UNKNOWN at the sixth token has no
valid table entry.

IMPLEMENTATION / EVIDENCE

compiler/parser/predictive.py

docs/evidence/final-analysis-20260927.json

Bounds protect the parser

The formal pushdown model is implemented with explicit resource limits.

256

Input tokens

1,000

Characters in one token

2,000

Recorded parse steps

1,024

Stack symbols

IMPLEMENTATION / EVIDENCE

`compiler/parser/predictive.py`

Four authored compiler classes, many functions

Model the actual implementation, not invented service objects.

compiler/lexer

tokenizer.py

Token: namedtuple with text and category.

compiler/lexer

reference.py

ReferenceEntry: immutable supplied row.

compiler/parser

grammar.py

GrammarError: explicit notation failure.

compiler/parser

analysis.py

_Transformations: bounded rewrite log.

IMPLEMENTATION / EVIDENCE

docs/diagrams/class-compiler.puml

compiler/parser/analysis.py

CS4110 / SETA / 27 SEP 2026

compiler/lexer/tokenizer.py

Shared storage is not an account

One protected workspace supports all public visitors.

PUBLIC POLICY

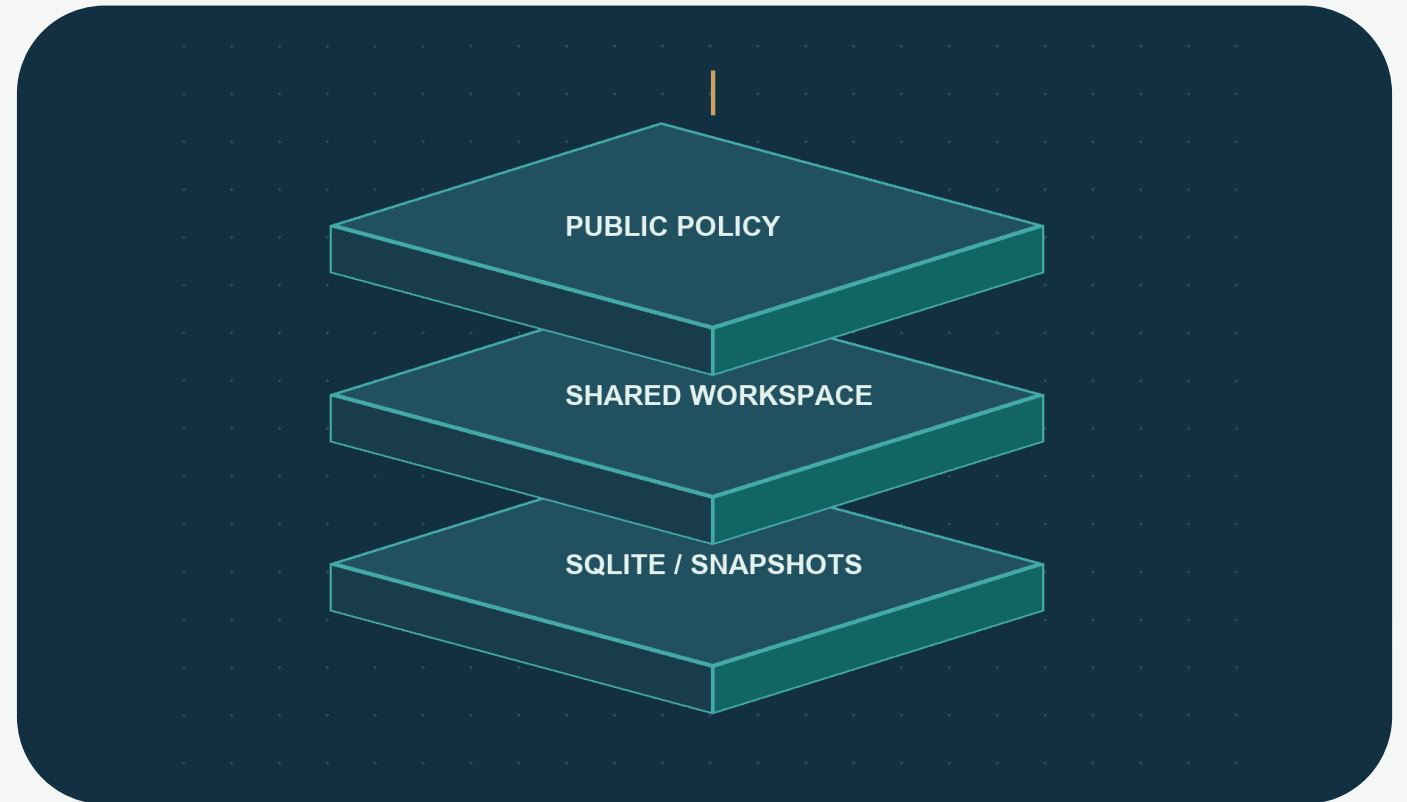
PublicWorkspaceStore narrows allowed writes.

SHARED STORE

SharedWorkspaceStore supplies common persistence.

SQLITE

workspace.sqlite3 holds shared records and test snapshots.



IMPLEMENTATION / EVIDENCE

backend/public_access.py
backend/workspaces.py
CS4110 / SET A / 27 SEP 2026

backend/shared_workspace.py

Save once, retry safely

The snapshot and request mapping commit together.

01

LOOKUP

Check an existing request ID



02

VALIDATE

Text and grammar must agree



03

COMPUTE

Build a complete snapshot



04

TRANSACTION

Save test + request mapping atomically

FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

backend/analyzer_history.py

backend/shared_workspace.py

Same retry versus a new stale request

Grammar changes must not destroy the meaning of idempotency.

01

IDENTICAL OLD RETRY

Can return its original snapshot after an offline grammar change.
It is not a new computation.

02

NEW STALE REQUEST

A new request with outdated grammar returns 409.
Refresh the shared grammar before trying again.

IMPLEMENTATION / EVIDENCE

backend/public_access.py
frontend/src/FrancAnalyzer.tsx
CS4110 / SET A / 27 SEP 2026

backend/analyzer_history.py

What is stored in a test?

The saved evidence is enough to explain its original result.

TEXT + GRAMMAR

The exact input and chosen rule snapshot.

LEXICAL RESULT

Raw tokens, categories, counts and clues.

PARSE RESULT

Accepted flag, consumed count, error and trace.

METADATA

Stable ID, recorded time and contextual summaries.

Web SQLite and desktop CSV are different

Avoid explaining the active public store with the wrong persistence helper.

01

PUBLIC WEB

SharedWorkspaceStore / WorkspaceStore.
Shared SQLite database.
Public tests append; reference data stays
read-only.

02

RETAINED DESKTOP

data_collector/dataset.py.
CSV with a lock and atomic replacement.
Local compatibility, not the public write
path.

IMPLEMENTATION / EVIDENCE

backend/shared_workspace.py
data_collector/dataset.py
CS4110 / SET A / 27 SEP 2026

backend/workspaces.py

Read-only rules live on the server

Hiding an edit button is not the enforcement mechanism.

Request family	Public result
Analyzer test create/read	Allowed with request protections
Collection and saved grammar reads	Allowed
Collection / grammar / recording writes	403
Account, workspace, import, private maintenance	404
Invalid Origin or CSRF on protected POST	403
More than 120 requests / 60 seconds / address	429 with Retry-After

IMPLEMENTATION / EVIDENCE

backend/public_access.py
backend/tests/test_public_access.py
CS4110 / SETA / 27 SEP 2026

backend/rate_limits.py

Recorded audio is retrieved, not generated

Two read paths serve existing media.

01

COLLECTION AUDIO

Play an audio attachment that already belongs to a record.

02

RECORDED READING

Look up a compatible text/language reading and stream its file.

03

HONEST ABSENCE

No recording means unavailable, not a synthesized substitute.

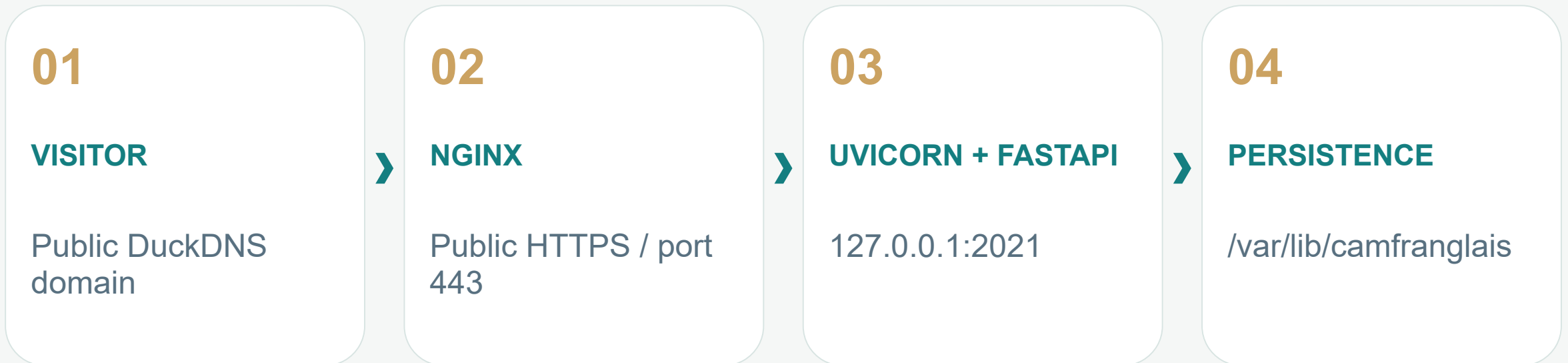
IMPLEMENTATION / EVIDENCE

backend/audio_api.py
frontend/src/RecordedReadings.tsx
CS4110 / SETA / 27 SEP 2026

backend/readings_api.py
frontend/src/voice.ts

HTTPS outside, port 2021 inside

camfranglais.duckdns.org uses a private loopback upstream.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

DOCUMENTED TOPOLOGY / loopback upstream, public HTTPS

IMPLEMENTATION / EVIDENCE

README.md
backend/config.py
CS4110 / SETA / 27 SEP 2026

backend/web.py
tools/run_app.py

Run and configure the application

The launcher checks prerequisites instead of hiding failures.

repository root

start.ps1

Convenient local Windows entry point.

tools

run_app.py

Validate launch settings and serve one origin.

backend

config.py

Environment, public URL and storage settings.

backend

main.py

Construct the FastAPI application.

backend

web.py

Serve the built frontend and web policy.

IMPLEMENTATION / EVIDENCE

start.ps1
backend/config.py
CS4110 / SETA / 27 SEP 2026

tools/run_app.py
backend/main.py

Recovery does not rewrite history

Use the failure message to choose the next safe action.

TIMEOUT

Recover an identical pending request before creating another.

STALE GRAMMAR

Refresh the saved grammar for a new computation.

BUSY STORAGE

Retry when the service is available.

OLD RESULT

Inspect the original snapshot; do not reinterpret it as a rerun.

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx
backend/analyzer_history.py
CS4110 / SETA / 27 SEP 2026

frontend/src/api.ts
backend/public_access.py

A fixed study, not a live benchmark

The evaluation uses the same twelve original strings.

12

Supplied originals

81

Tokens retained

10

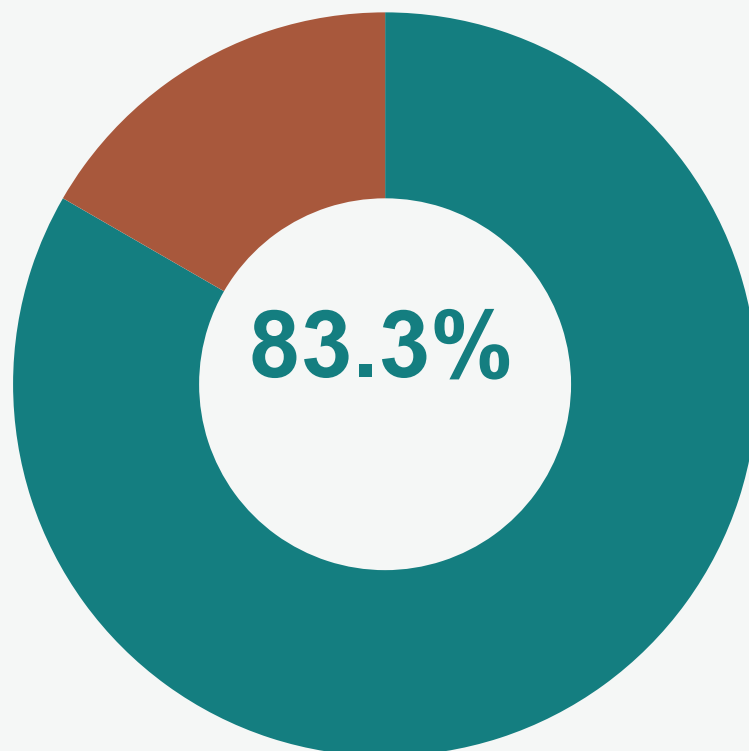
Grammar accepts

2

Explicit grammar rejects

Ten of twelve fit the saved grammar

83.3% category-grammar match is not linguistic accuracy.



ACCEPTED

10 original statements

REJECTED

2 originals with unresolved sixth tokens

INTERPRETATION

A result for this corpus and these rules only

Boundary cases expose the rules

Seven constructed controls supplement the twelve originals.

Control	Constructed input	Saved-grammar result
C01	(empty)	REJECT
C02	school	REJECT
C03	je	REJECT
C04	je suis kass.	ACCEPT
C05	je suis kass?!	REJECT
C06	je je suis	REJECT
C07	je suis alli	REJECT

IMPLEMENTATION / EVIDENCE

docs/evidence/final-analysis-20260927.json

compiler/tests/test_yaounde_grammar.py

Evidence has a clear boundary

State what is measured and what still needs human confirmation.

01

VERIFIED

Token outputs, grammar transformations, traces and saved-data behavior.

02

QUALIFIED

Supplied glosses and lexical annotations need linguistic review.

03

UNCONFIRMED

Original fieldwork dates, collectors, speakers and consent.

Tests check different layers

A successful build is not the only form of evidence.

COMPILER

Exact tokenization, grammar controls and trace transitions.

API + STORAGE

Public permissions, snapshots and safe retries.

FRONTEND

Page behavior, request states and visible results.

DOCUMENTS

Diagrams, source coverage and reproducible report evidence.

IMPLEMENTATION / EVIDENCE

compiler/tests/test_yaounde_grammar.py
frontend/tests/unit/FrancAnalyzer.test.tsx
CS4110 / SETA / 27 SEP 2026

backend/tests/test_public_access.py
tools/tests/test_report_evidence.py

File map: UI and the request path

A compact index for questions during the demonstration.

frontend/src

FrancAnalyzer.tsx

Where does Analyze begin?

frontend/src

TestStatistics.tsx

Where are history metrics displayed?

frontend/src

Collection.tsx

Where are records browsed?

frontend/src

api.ts

Where are errors and headers handled?

backend

analyzer_api.py

Where does the request enter?

backend

analyzer_history.py

Where is its snapshot managed?

IMPLEMENTATION / EVIDENCE

frontend/src/FrancAnalyzer.tsx

backend/analyzer_api.py

CS4110 / SETA / 27 SEP 2026

frontend/src/api.ts

backend/analyzer_history.py

File map: compiler and persistence

The computational core stays separate from the public interface.

compiler/lexer

tokenizer.py

How are raw tokens and labels produced?

compiler/parser

analysis.py

How are grammar sets and tables calculated?

compiler/parser

predictive.py

How does the stack accept or reject?

compiler/parser

yaounde.py

Why were these grammar rules chosen?

backend

shared_workspace.py

How are public requests saved once?

backend

workspaces.py

Where is the SQLite workspace implemented?

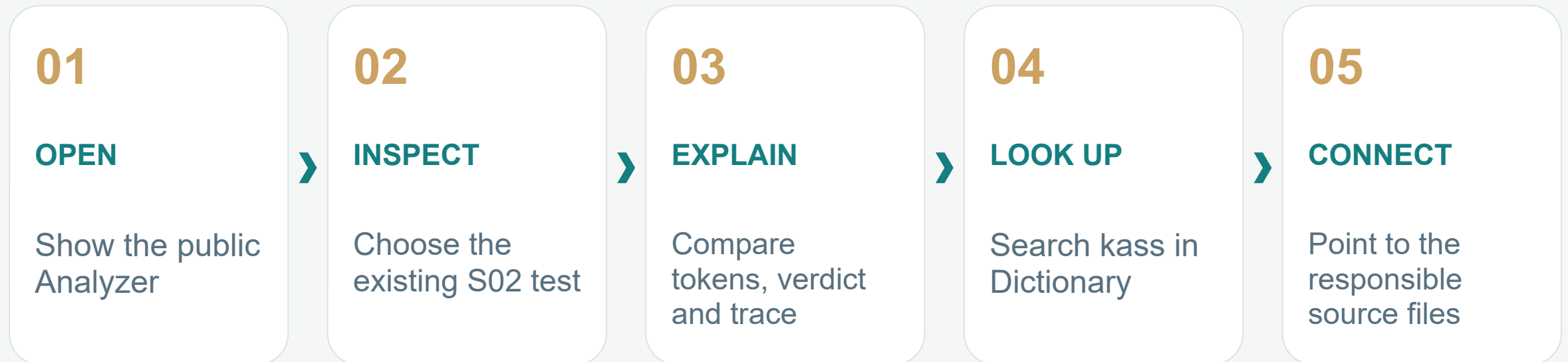
IMPLEMENTATION / EVIDENCE

compiler/lexer/tokenizer.py
compiler/parser/predictive.py
CS4110 / SETA / 27 SEP 2026

compiler/parser/analysis.py
backend/shared_workspace.py

A safe demonstration in five moves

Use an existing saved result for a read-only rehearsal.



FOLLOW THE DATA / KEEP EACH RESPONSIBILITY VISIBLE

IMPLEMENTATION / EVIDENCE

frontend/src/App.tsx
frontend/src/Dictionary.tsx
CS4110 / SETA / 27 SEP 2026

frontend/src/FrancAnalyzer.tsx
docs/README.md

Present the story; keep the depth

All 100 slides are included. A short oral route is optional.

OPEN / 01, 05

Introduce the product and its five pages.

DEMO / 13, 17, 24, 35

Show a request, a result, two verdicts and lookup.

EXPLAIN / 56, 73, 88

Connect lexer, parser and deployment.

CLOSE / 100

Return to the value and invite questions.

From language to evidence.

Your questions, traced to the code.

PRESERVE

The original text

EXPLAIN

The rules and trace

RETAIN

The original result

camfranglais.duckdns.org